



J. Dairy Sci. TBC

<https://doi.org/10.3168/jds.2025-27833>

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Standardizing case definitions for hoof lesions and lameness: A scoping review to improve machine learning applications in dairy cattle

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ABSTRACT

Lameness in dairy cattle, primarily caused by hoof lesions, remains a widespread issue in North America with significant welfare, economic, and sustainability impacts. Although emerging technologies incorporating machine learning offer promising solutions for the detection of hoof lesions and lameness, inconsistent outcome definitions for hoof lesions and lameness may limit model generalizability and on-farm application. This scoping review aims to examine how hoof lesions and lameness outcomes are defined and classified across studies, identify inconsistencies in outcomes definitions, and provide guidance for improving standardization to improve the effectiveness of machine learning applications. A systematic search was conducted in October 2024 across 4 databases: Scopus, PubMed, Agricola, and CAB Digital Library. Of the 1,149 identified references, 1.7% (20/1149) of studies met the inclusion criteria and were selected for data extraction. Among these, 55% (11/20) developed outcomes based on locomotion scores, 35% (7/20) used hoof lesion data, and 10% (2/20) incorporated both. Of the studies using only locomotion scores ($n = 12$), 66.7% (8/12) used a 5-point scale, 16.7% (2/12) used a 4-point scale, and 16.7% (2/12) used a 3-point scale. Of these, 50% (6/12) created binary outcomes, 33.3% (4/12) created 3-level outcomes, and 8.3% (1/12) used 5-level outcomes. Among the studies using hoof lesion outcomes ($n = 8$), 62.5% (5/8) assessed multiple lesion types, while 37.5% (3/8) focused on differentiating stages of digital dermatitis. Additionally, 37.5% (3/8) did not provide any descriptions of the hoof lesions used. Descriptions of the hoof lesions also varied among those that did provide criteria. Reporting of population characteristics was limited: country of data collection was reported in 95% (19/20) of studies, breed(s) in 70% (14/20), and lactation-related information in 65% (13/20). Bedding type, flooring, and

outdoor access were reported in 30% (6/20), and production system type in only 10% (2/20). Furthermore, only 10% (2/20) of studies validated their models using data from herds external to the original study population. To improve the generalizability and real-world utility of machine learning applications in dairy cattle management, future research must provide clear outcome definitions, detailed reporting of study and population characteristics, perform external validation, and evaluate the impact of outcome classification.

Key words: mobility classification, precision livestock farming, predictive modeling

INTRODUCTION

In North America, lameness prevalence in dairy herds is estimated to range between 10% and 55% (von Keyserlingk et al., 2012; Cook et al., 2016; Adams et al., 2017), making it a significant concern for dairy production. Lameness is commonly characterized by abnormal gait or posture resulting from pain, most often caused by hoof lesions (Cramer and Solano, 2024), which are broadly categorized as infectious or noninfectious (Greenough, 2007). Lameness compromises animal welfare by causing pain (Whay et al., 1998), disrupting natural behaviors (Magrin et al., 2023), and reducing both milk production (Dolecheck et al., 2019; Puerto et al., 2021) and reproductive efficiency (Mellado et al., 2018). Economically, lameness results in treatment and labor costs estimated to range between \$100 and \$1,000 per case (Ózsvári, 2017; Dolecheck et al., 2019), alongside losses due to reduced BW, impaired reproductive performance, and increased culling (Liang et al., 2017; Ózsvári, 2017). Environmentally, lameness indirectly contributes to higher GHG emissions via premature culling (Dallago et al., 2021) and reduced feed efficiency (Chen et al., 2016; Mostert et al., 2018). Given the societal concerns about the ethical treatment of production animals (Fraser, 2008), addressing lameness is essential not only for cow health and farm profitability but also for broader environmental and social sustainability.

Received October 26, 2025.

Accepted January 15, 2026.

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The list of standard abbreviations for JDS is available at adsa.org/jds-abbreviations-26. Nonstandard abbreviations are available in the Notes.

Given these negative effects of lameness, reliable detection of hoof lesions and lameness is critical for timely intervention and effective management. Currently, the primary on-farm method for identifying lame cows is passive surveillance, performed during other routine farm tasks. However, this approach typically identifies only severe cases, leading producers to underestimate lameness prevalence (Leach et al., 2010). Additionally, differing perceptions of lameness also vary among producers, veterinarians, researchers, and farm advisors and can result in delayed treatment (Knauss et al., 2022). Identifying cows that would benefit from hoof trimming and treatment is further complicated by differences in locomotion as a result of hoof lesion type (Flower and Weary, 2006; Blackie et al., 2013) and severity (Frankena et al., 2009; Klawitter et al., 2019). These challenges highlight the need for more advanced and objective methods to improve lameness detection and management.

Emerging technologies incorporating machine learning offer scalable and accurate alternatives to human assessments by addressing their challenges. However, their implementation relies on the consistent and accurate classification of hoof lesions and lameness, which requires standardized case definitions (PAHO, 1999). Scoring systems for lameness vary, as reviewed by Schlageter-Tello et al. (2014), using continuous scales (e.g., 0–1, 0–10, and 0–100) to ordinal scales (e.g., lame or not lame). For hoof lesions, methods such as the 13-zone method (Shearer et al., 2004), International Committee for Animal Recording (ICAR) Claw Atlas (Egger-Danner et al., 2015), and severity scoring systems are used for white line disease and sole ulcers (Leach et al., 1997; Miguel-Pacheco et al., 2017), and for digital dermatitis (Döpfer et al., 1997; Krull et al., 2014). However, variations in these systems create inconsistencies across studies, limiting the ability of machine learning models to generalize and perform reliably across datasets and environments.

This scoping review explores the case definitions and classification criteria for hoof lesions and lameness, aiming to identify discrepancies in standardization and provide insights for improving the consistency and applicability of machine learning models. In addition to evaluating outcome definitions, the review also summarized key study design and population characteristics and reported evaluation metrics to provide an overview of methodological approaches used across studies and to offer considerations for future research.

MATERIALS AND METHODS

Protocol and Registration

To initiate the scoping review, an a priori protocol was established in accordance with the methodological

guidelines outlined in the Preferred Reporting Items for Systematic Reviews and Meta-Analyses Extension for Scoping Reviews (Tricco et al., 2018). The methodological framework proposed by Arksey and O'Malley (2005) was used, along with recommendations from Levac et al. (2010) and Peters et al. (2015). Furthermore, the application of these frameworks to veterinary medicine was informed by the work of Sargeant and O'Connor (2020), which recommends adapting the population, intervention, comparisons, and outcome framework specifically for systematic reviews. Our methodology reflects the differentiation made by Sargeant and O'Connor (2020), where systematic reviews concentrate on specific comparisons, while scoping reviews systematically chart the existing research on a broader subject without the extraction of detailed study results.

Therefore, in our scoping review, the population consists of lactating dairy cows, the intervention or comparison focuses on various machine learning models applied in studies to identify hoof lesions and lameness, and this review is intended to describe how the outcomes (hoof lesions and lameness case definitions) are described, as well as to summarize the machine learning algorithms used, the evaluation metrics applied, and the study design and population characteristics presented across studies.

Eligibility Criteria. Articles were eligible for inclusion if they were a primary research study and aimed at developing a machine learning algorithm that was applied to lactating dairy cows with the aim of classifying hoof lesions and lameness. Additionally, articles were only eligible to be included if the full article was available online, written in English, and had at least 500 words; there were no restrictions on the location or date of publication. This approach aligns with the inclusion criteria used in similar scoping reviews, such as those by McFarlane et al. (2022) and Wilson et al. (2023).

Information Sources and Search Strategy. The literature search was conducted across 4 databases: Scopus (Elsevier), PubMed (NIH National Library of Medicine), Agricola (USDA National Agricultural Library), and CAB Digital Library (CABI). The search was applied to titles, abstracts, and keywords on October 9, 2024. The search string combined terms for hoof (Hoof OR Claw OR Foot), lameness (Lame* OR Lesion OR Locomotion OR Mobility OR Defect), cattle (Cow* OR Cattle), and machine learning (“Machine Learning” OR class* OR predict* OR “Artificial Intelligence”) using the Boolean operator “AND.” Keywords were initially developed by DS and GC, with refinement assistance from the veterinary medicine subject librarian (AN).

Selection of Evidence. The research article screening process is shown using a flowchart (Figure 1). All search results from each database were uploaded into a web-based collaboration software platform that streamlines

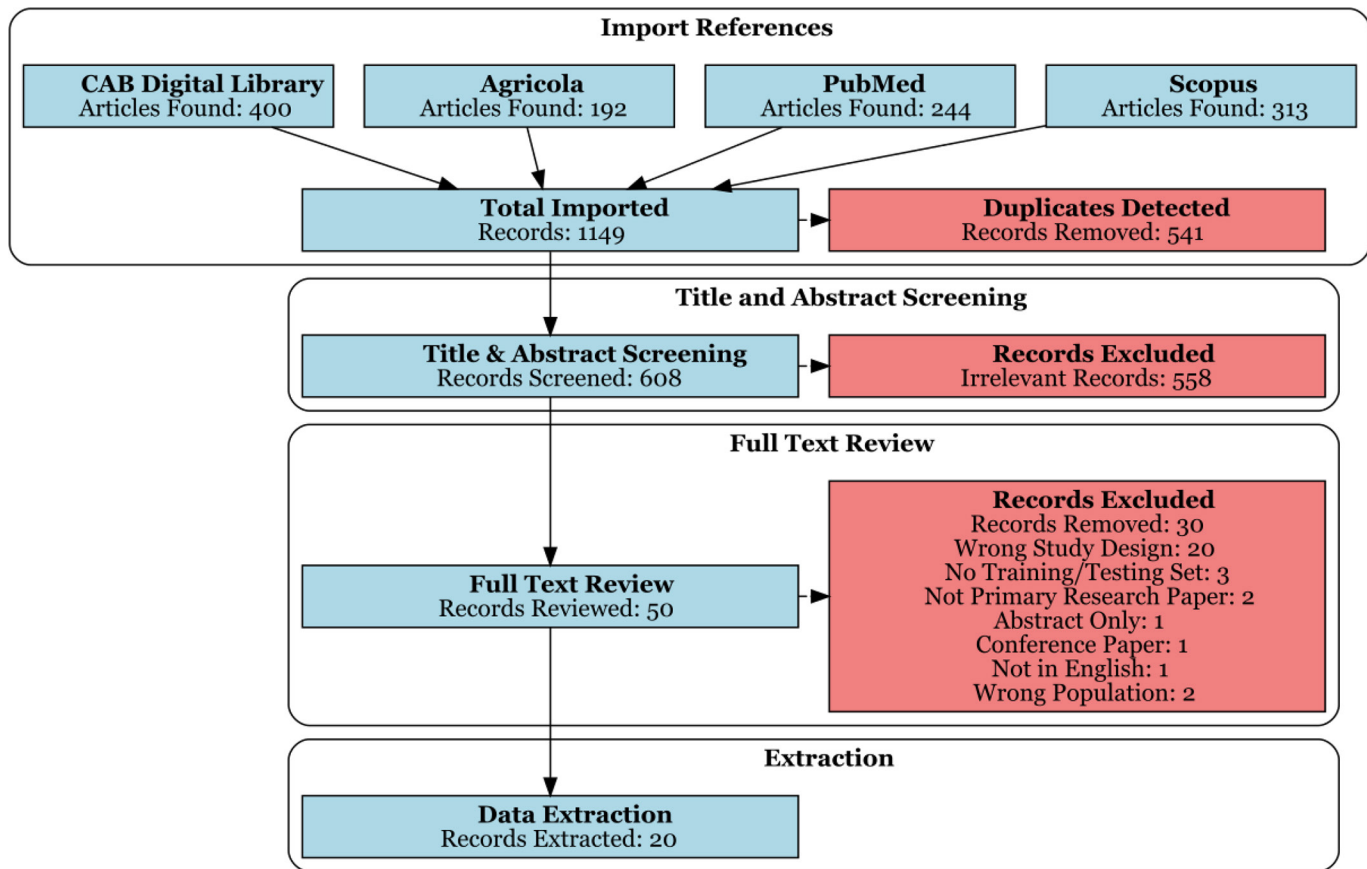


Figure 1. The flow diagram of study selection. The diagram outlines the study selection process in 4 main stages: (1) importing references from 4 databases, (2) title and abstract screening, (3) full-text review, and (4) data extraction. At each stage, the number of records removed or excluded is shown, along with the reasons for exclusion where applicable.

the production of systematic and other literature reviews (Covidence; Veritas Health Innovation, Melbourne, Australia, available at www.covidence.org). In the initial stage, search results were combined and deduplicated.

The review process then proceeded into 2 screening phases. The first phase was the title and abstract screening, which was independently conducted by 2 reviewers (DS and GC) to assess relevance. Discrepancies were resolved through discussion and consensus. In the second phase, a full-text review of the remaining articles was conducted, again independently conducted by DS and GC, with any disagreements similarly resolved through discussion. The following questions were used to assess eligibility:

- Is the article available online and at least 500 words?
- Does the article describe a primary research study?
- Did the study develop a machine learning algorithm or model applied to lactating dairy cattle?
- Did the study use a training/testing set?

- Did the study aim to classify hoof lesions and lameness in lactating dairy cattle?
- Does the study describe a definition for lameness or hoof lesion (scoring system, size, cites another study as a reference)?

Data Charting Procedure and Items. For all studies that met inclusion criteria following the full-text review, data were independently extracted by DS and GC using the software platform. Any discrepancies were resolved through discussion and consensus. A data extraction template was developed to guide the extraction process.

For each study, information was recorded on the total number of observations in the training and testing sets and whether the testing data originated from the same dataset as the training data. It was also noted whether the study employed cross-validation methods.

The study outcome was categorized as locomotion scores, hoof lesions, or both. Details were recorded on who assessed the outcome (evaluator; e.g., trained evaluator or veterinarian) and how many evaluators were

involved. The outcome criteria were documented, including whether lameness was assessed using a numerical scale, the specific type of hoof lesion assessed, and how the outcome was described or referenced in the text.

For studies reporting locomotion scores, the specific locomotion score, description, and assessment criteria were extracted. For hoof lesion outcomes, the lesion name and assessment criteria were documented. Additionally, it was noted whether the study accounted for the chronicity of lameness or the recurrence of hoof lesions.

Additional information on study design and herd characteristics was recorded, including whether the training and testing sizes were explicitly reported, the geographic location of the study, and how lactation distribution was reported. Housing and environmental details, including bedding type, outdoor access, flooring type, and production system, were also extracted. Finally, the name(s) of the machine learning algorithms used in the study were recorded, along with the evaluation metrics used to assess algorithm performance.

Synthesis of Results. Information was summarized for each included study, including the type of outcome assessed, the individual(s) responsible for recording the outcome, the number of evaluators involved, the outcome criteria used, and any citations referencing those criteria. The total number of citations referencing locomotion scoring scales and hoof lesion definitions was calculated, with corresponding score descriptions and assessment criteria extracted for locomotion references, and hoof lesion names and assessment criteria documented for references. Study design and population characteristics were also summarized, along with the frequency of machine learning algorithms used and the evaluation metrics reported to assess model performance. A risk of bias assessment was not conducted, as scoping reviews are intended to map the breadth of existing literature rather than evaluate methodological quality, and thus typically do not include critical appraisal (Tricco et al., 2018).

RESULTS AND DISCUSSION

A total of 1,149 articles were retrieved from the databases Agricola, CABI, PubMed, and Scopus. The progression of studies through each stage of the screening process is shown in Figure 1. After removing 47% (541/1,149) of the articles that were duplicates, 53% (608/1,149) of the articles remained for title and abstract screening. Out of the remaining articles, 92% (558/608) were excluded due to irrelevance, leading to 8% (50/608) of the articles proceeding to the full-text review stage. During the full-text review, 60% (30/50) of the articles were excluded, primarily due to wrong study design, specifically not applying a machine learning approach. After the full-text review, 40% (20/50) of the studies satisfied

the inclusion criteria and were incorporated into the data extraction phase.

Study Population Characteristics

Herd- and farm-level characteristics from the 20 studies are summarized in Tables 1 and 2, respectively. Among these characteristics, country of data, breeds used, and lactation status were the 3 most frequently reported (Figure 2). Most of the studies that reported these characteristics used data collected in the United Kingdom or United States (Figure 3), focused on Holstein/Friesian cows (Figure 4), and categorized lactation status as either primiparous or multiparous (Figure 5). Aravamuthan et al. (2024a) was the only study to include data from multiple countries, although Lemmens et al. (2023) and Mineur et al. (2017) were the only ones to use data from multiple breeds.

The reporting of the population characteristics was inconsistent, as 5% (1/20) did not specify the country where the data were collected (Figure 2). Similarly, 30% (6/20) of studies did not report the breed of cows included, and 35% (7/20) did not describe lactation status. Without transparency in population characteristics, it becomes difficult to determine the production systems, breeds, or management contexts to which these models may apply, thereby limiting their adoption and utility outside of the original study population.

Study Design Characteristics Reported

Details related to machine learning methodology across the reviewed studies are presented in Table 3. Although 25% (5/20) of studies did not fully specify the size of their datasets, all studies provided enough information to determine whether they used a single dataset or multiple distinct datasets. Specifically, 60% (12/20) used a single dataset (e.g., from 1 farm or time period), whereas 40% (8/20) used multiple datasets (e.g., from different farms or multiple subsets from a larger dataset).

Regarding validation strategies, 55% (11/20) of studies used only a training/testing split, whereas 40% (8/20) used cross-validation methods, and 5% (1/20) incorporated both approaches (Figure 6). Among those that used a training/testing approach and reported split details, training sets ranged from 50% to 90% of the total dataset. Meanwhile, studies that used cross-validation applied techniques such as leave-one-out, 5-fold, or 10-fold cross validation. One study, Altay and Delialioğlu (2022), did not report how the dataset was partitioned for validation. Only Post et al. (2021) and Zhou et al. (2023) evaluated their models using external data that were collected from farms different from those used for training.

Table 1. The herd-level characteristics from the 20 studies included in the review, including the breed(s) of cattle used, how lactation stage was defined or categorized, and the country where the data originated

Study	Breed included in study	Lactation description	Countries included in study
Altay and Delialioğlu, 2022	Simmental	Average lactation	Turkey
Anagnostopoulos et al., 2021	Holstein	Primiparous/multiparous	United Kingdom
Aravamuthan et al., 2024a	Not stated	Not stated	Canada, the Netherlands, United States
Aravamuthan et al., 2024b	Not stated	Not stated	United States
Barden et al., 2021	Holstein	Lactation number	United Kingdom
Busin et al., 2019	Holstein-Friesian	Not stated	United Kingdom
Coşkun et al., 2023	Brown Swiss	Average lactation	Turkey
He et al., 2022	Not stated	Not stated	United Kingdom
Kang et al., 2020	Holstein	Primiparous/multiparous	China
Kang et al., 2022	Holstein	Primiparous/multiparous	China
Lavrova et al., 2023	Not stated	Lactation number	Germany
Lemmens et al., 2023	Brown Swiss, Fleckvieh, Holstein and Red Friesian, mixed	Not stated	Austria
Mineur et al., 2017	Brown Swiss, Fleckvieh, Holstein	Lactation number	Austria
Ghotoorlar et al., 2012	Not stated	Not stated	Not stated
Neupane et al., 2024	Holstein	Primiparous/multiparous	United States
Pastell and Kujala, 2007	Not stated	Primiparous/multiparous	Finland
Post et al., 2021	German Holstein	Average lactation	Germany
Volkman et al., 2021	German Holstein	Lactation number	Germany
Zhao et al., 2023	Holstein	Not stated	United States
Zhou et al., 2023	Holstein	Primiparous/multiparous	China

This lack of external testing raises concerns about the generalizability of model performance to new or independent settings, as internal validation alone may not accurately reflect future predictions (Bleeker et al., 2003). Models developed using data from a single farm may be affected by sample selection bias, where training data are correlated with characteristics unique to that farm, reducing the likelihood of successful generalization. Likewise, covariate shift may occur when the distribution of pre-

dictor variables in the target population differs from that of the training data (Gu and Oelke, 2019). These issues are particularly concerning when models are trained on single-source populations (Yu and Eng, 2020), as they may underperform when applied to more diverse or differently managed herds. As machine learning becomes more widely used in animal health applications, especially in commercial contexts, future research should

Table 2. The farm-level characteristics from the 20 studies included in the review, including production system type, flooring and bedding materials, and information on outdoor access when reported, highlighting differences in housing and management practices

Study	Production system	Flooring type	Bedding type	Outdoor access
Altay and Delialioğlu, 2022	Not stated	Not stated	Not stated	Not stated
Anagnostopoulos et al., 2021	Not stated	Not stated	Not stated	Not stated
Aravamuthan et al., 2024a	Not stated	Not stated	Not stated	Not stated
Aravamuthan et al., 2024b	Not stated	Not stated	Not stated	Not stated
Barden et al., 2021	Not stated	Grooved concrete	Cubicles deep bedded with sand	Housed year round
Busin et al., 2019	Not stated	Not stated	Not stated	Not stated
Coşkun et al., 2023	Not stated	Not stated	Not stated	Not stated
He et al., 2022	Not stated	Slatted	Sand-bedded cubicles	Continuously housed
Kang et al., 2020	Not stated	Not stated	Not stated	Not stated
Kang et al., 2022	Not stated	Not stated	Not stated	Not stated
Lavrova et al., 2023	Not stated	Not stated	Not stated	No access to pasture
Lemmens et al., 2023	9: Conventional; 1: Organic	4: Concrete; 3: Concrete w/ rubber; 2: Slatted; 1: Partly slatted/concrete	8: Straw for cubicle bedding; 2: Separated manure w/ 1 offering compost area	1: Pasture for cows
Mineur et al., 2017	Not stated	Not stated	Not stated	Not stated
Ghotoorlar et al., 2012	Not stated	Not stated	Not stated	Not stated
Neupane et al., 2024	Organic	Concrete slatted	Sand bedding	Access after April 23
Pastell and Kujala, 2007	Not stated	Hard, slatted	Rubber mats	Not stated
Post et al., 2021	Not stated	Not stated	Not stated	Not stated
Volkman et al., 2021	Not stated	Not stated	Not stated	Not stated
Zhao et al., 2023	Not stated	Not stated	Not stated	Not stated
Zhou et al., 2023	Not stated	Concrete solid w/ sawdust	Cubicles	Access to exercise yard

Swartz et al.: LAMENESS LESION CASE DEFINITIONS

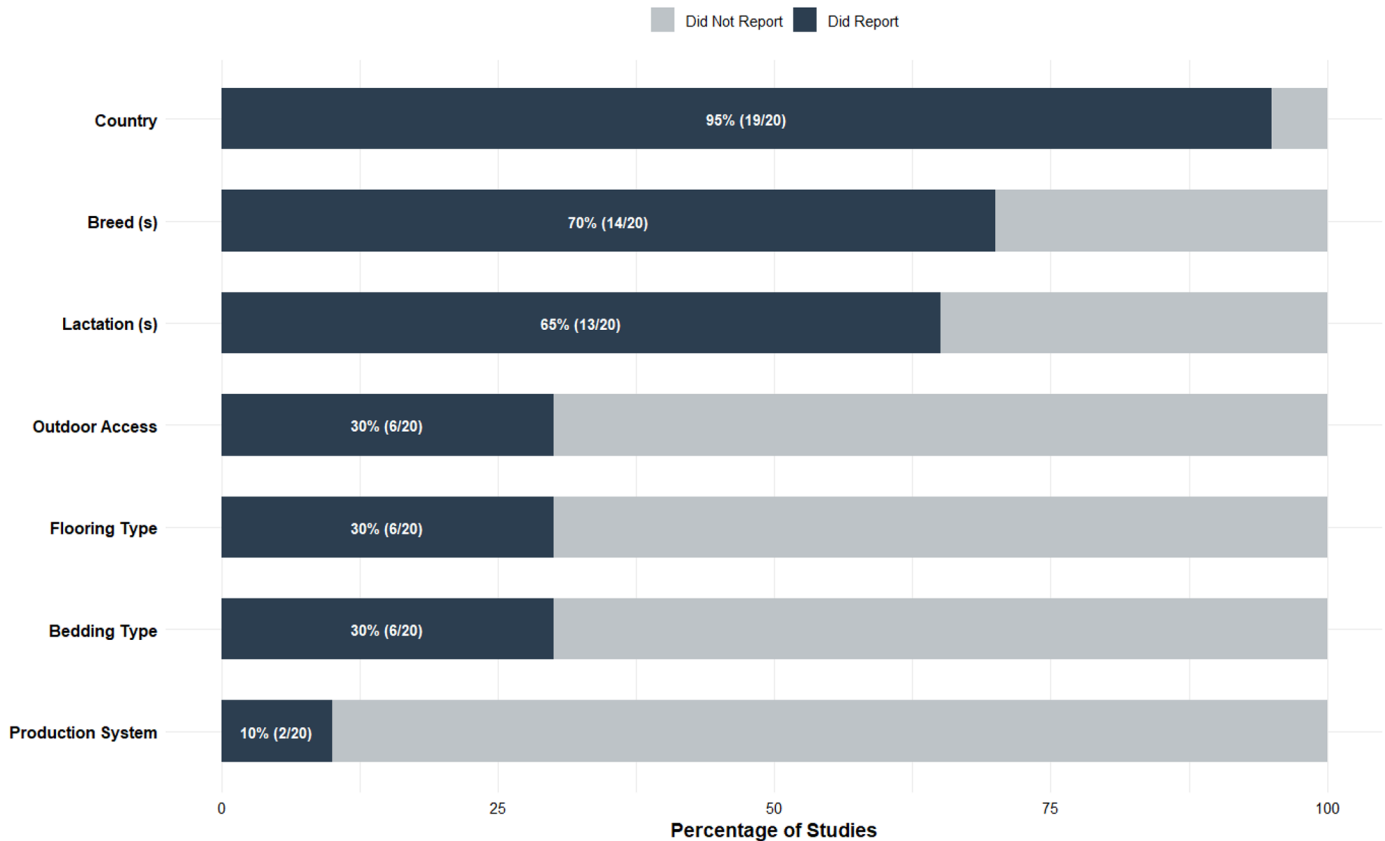


Figure 2. The percentage of studies (n = 20) that reported specific herd- and farm-level characteristics. Each characteristic is listed on the y-axis, with the x-axis representing the percentage of studies.

prioritize external validation using data from herds not involved in development.

Model Evaluation and Reporting

The most commonly used machine learning algorithm was random forest, followed by logistic regression (Figure 7). As reviewed by Sarker (2021), random forest is an ensemble method that constructs multiple decision trees using randomly selected subsets of data and features, combining their outputs to improve predictive accuracy and reduce overfitting. Logistic regression, in contrast, is a probabilistic model that estimates the likelihood of an outcome using a logistic function. Both algorithms are widely used for classification tasks and are suitable for categorical outcomes such as those used in classifying locomotion scores and hoof lesions.

Model performance was most often assessed using classification metrics, with specificity, sensitivity, and accuracy being the most frequently reported across studies (Figure 8). However, inconsistencies in terminology were common, as several studies reported functionally equivalent metrics under different names. For instance,

sensitivity, recall, and true positive rate all refer to the proportion of actual positives correctly identified by the model. Similarly, precision and positive predictive value both describe the proportion of predicted positives that are truly affected. Error proportion and misclassification rate both represent the percentage of incorrect predictions and are equivalent to one minus the model's overall accuracy. This inconsistent use of terminology can limit comparison across studies and reduce clarity in the interpretation of results.

Despite the range of metrics reported, the lack of standardization in terminology and reporting highlights a clear need for consistency across future studies. In particular, the animal and veterinary science fields may benefit from aligning their performance metrics with those commonly used in epidemiology, including accuracy, sensitivity, specificity, positive predictive value, and negative predictive value, which are widely used for evaluating diagnostic test performance and disease surveillance (Noah, 2024). Their consistent use would improve the clarity, comparability, and practical utility of machine learning model evaluations in livestock health applications.

Swartz et al.: LAMENESS LESION CASE DEFINITIONS

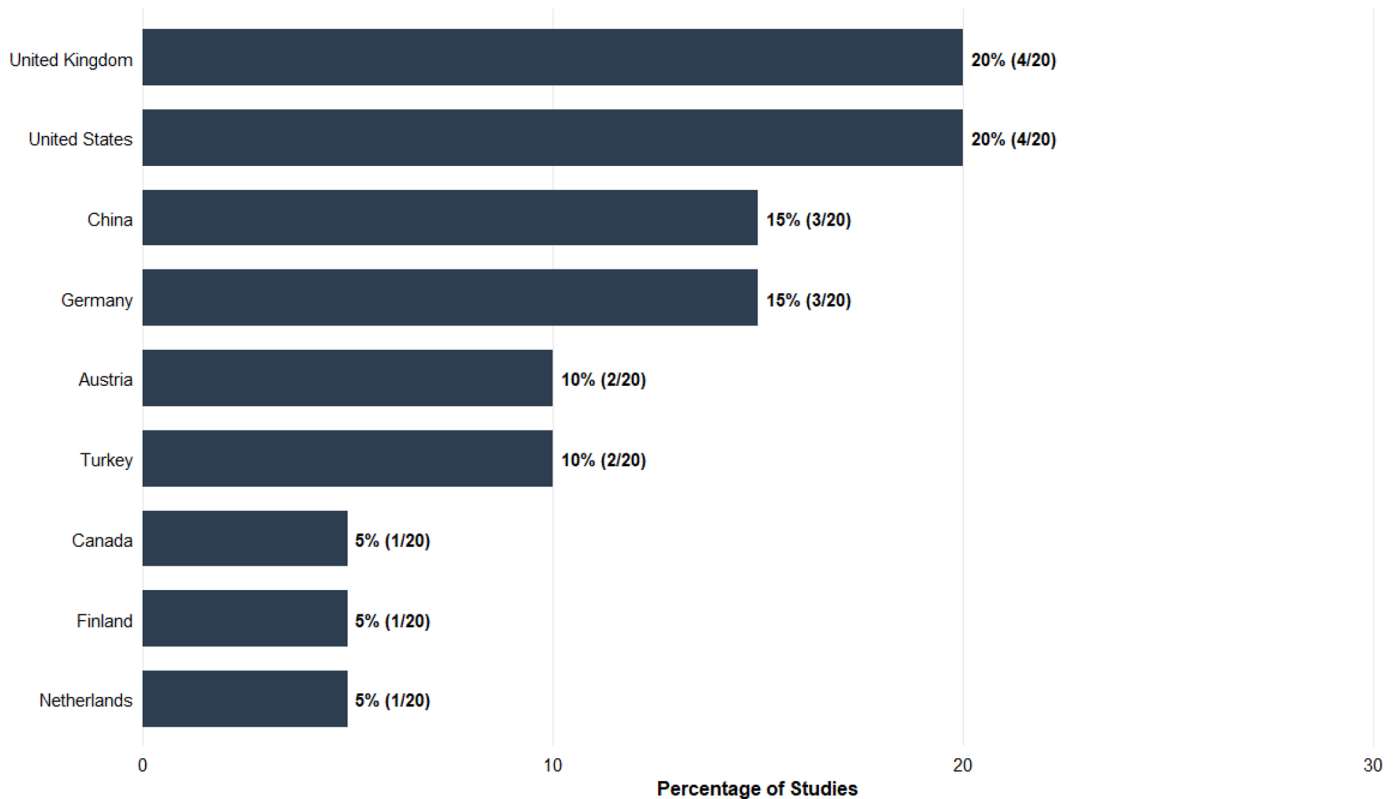


Figure 3. The percentage of included studies ($n = 20$) that used data in each country, with countries listed on the y-axis and the corresponding percentage on the x-axis. Each bar is labeled to the right with the percentage of studies and the proportion of studies that reported using data from that country.

A smaller number of studies also included agreement or repeatability metrics, such as Cohen's Kappa and Pearson correlation coefficient, especially when assessing consistency between automated predictions and human evaluators. Additionally, frames per second was occasionally reported to measure model speed, an important factor for real-time applications that use computer vision. Although these metrics can provide additional insight into model behavior, they were less commonly reported and secondary to classification performance measures. Importantly, many of the classification and repeatability evaluations rely on comparisons to human-labeled outcomes, highlighting the role of evaluators in shaping model development and assessment.

Evaluators

Across the studies included in this review, there was considerable variation in who performed locomotion scoring and hoof lesion classification, as well as in how these evaluators were described (Table 4). Evaluator selection and reporting are important considerations when evaluating the generalizability of machine learning models, as the individuals responsible for outcome classification

influence the quality of the data. Study personnel were the most commonly reported evaluator group, followed by veterinarian(s) and hoof trimmer(s; Figure 9). Study personnel included any mention of "Expert," "Assessor," "Observer," "Investigator," "Research Group," "Technician," or "Personnel." However, it is possible that some individuals labeled as veterinarians or hoof trimmers were part of the research team and may have been more accurately categorized as study personnel.

Because evaluators determine the outcome labels used for model training and testing, inconsistent or biased labeling may introduce label bias (van Giffen et al., 2022). Although this risk exists for all evaluator types, it may be increased among unblinded study personnel or those with an interest in the algorithm's performance. This is particularly relevant when outcomes are subjective rather than objective (Page et al., 2016). It has been shown in an RCT study that assessors who were blinded to treatment allocation were still able to correctly guess treatment groups when providing subjective outcome evaluations (van der Ende et al., 2023). Applying this to machine learning, if evaluators are the same people who are developing the algorithm, then they may adjust outcome classifications to align with expected model performance.

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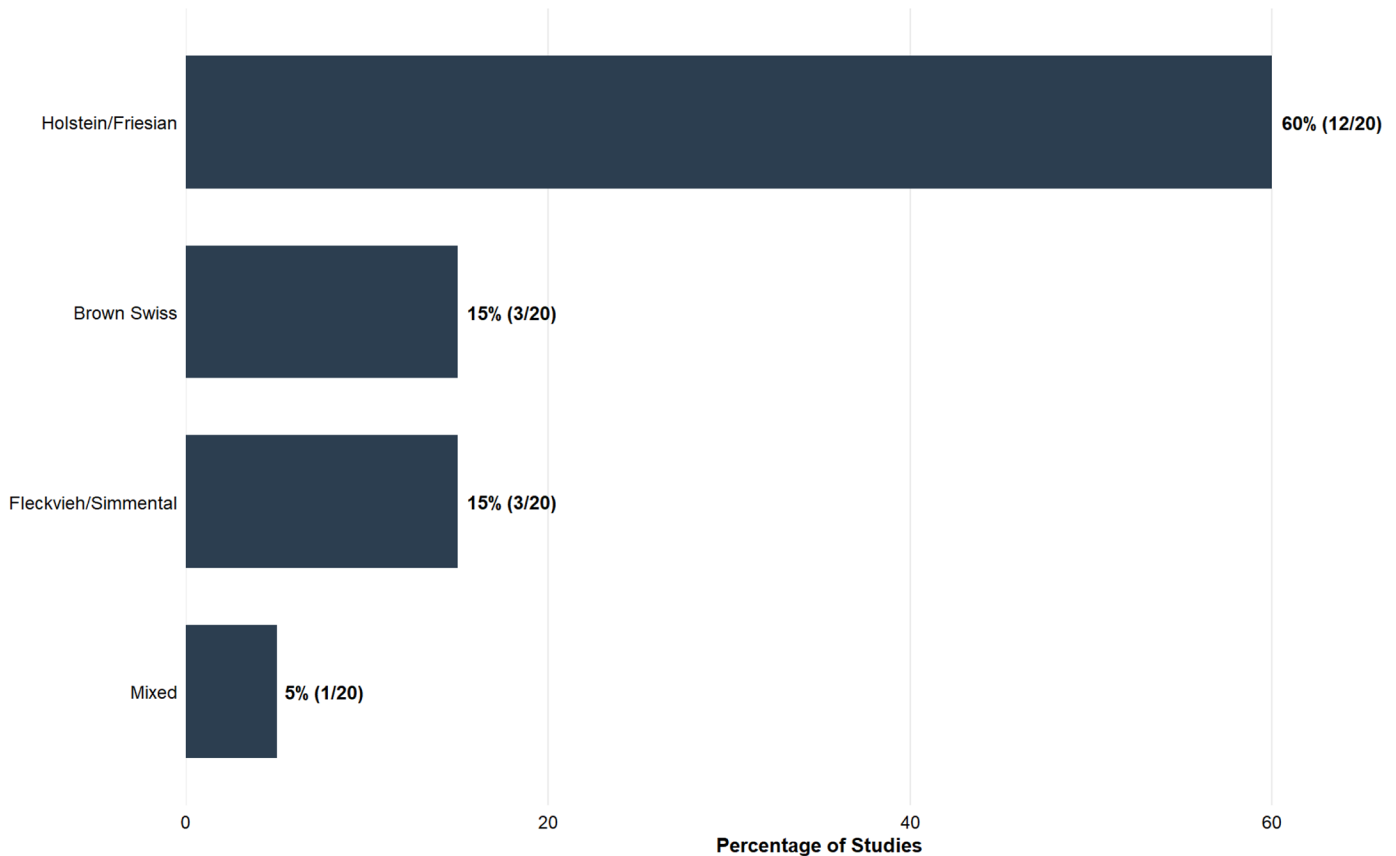


Figure 4. The percentage of studies that included each cattle breed. Breeds are listed on the y-axis, with the x-axis representing the percentage of included studies ($n = 20$) that used each breed. Bars are labeled to the right with the percentage and the proportion of studies that reported that breed.

To improve consistency, typically study personnel are trained together, and their outcome classifications may demonstrate high inter-rater agreement (Thomsen et al., 2008). However, in real-world applications, users of the algorithm may not receive the same training or apply classifications with the same consistency. This discrepancy can also result in label bias, which reduces the model performance and limits the perceived applicability of the tool. Therefore, transparent reporting of who the evaluators are, their training, and their connection to the study is essential for interpreting model validity and real-world relevance.

In addition to variation in the roles of evaluators, many studies lacked transparency regarding the number of individuals involved in scoring or classification. More studies failed to clearly state the number of evaluators involved than those that did (Figure 10). Clear reporting on the number of evaluators is important, as the choice between a single or multiple evaluators can affect both data quality and model generalizability. Using a single evaluator eliminates interobserver variability, which can improve consistency, but may introduce individual bias and limit the generalizability of outcomes. In contrast,

multiple evaluators better reflect real-world variability, potentially improving the generalizability of model findings. However, several studies showed a lack of clarity in reporting the number of evaluators.

The lack of clarity typically fell into 3 categories. First, some studies used vague language that implied the involvement of multiple evaluators but did not provide a definitive count. For example, some studies provided vague descriptions that suggested multiple evaluators but did not explicitly report how many. Lemmens et al. (2023) stated that “employees of the LKV visited each farm” and noted that “farms were visited twice by the first author,” creating uncertainty about how many LKV employees participated in data collection. Similarly, Lavrova et al. (2023) reported, “We used the scoring by veterinarians,” without clarifying the number of veterinarians involved. Pastell and Kujala (2007) wrote, “All cows were observed weekly for lameness during a normal gait by experienced personnel,” implying the involvement of multiple observers but without clear details. This uncertainty was further supported by their note that lameness “was not assessed daily by the observers,” suggesting that more than one individual conducted the

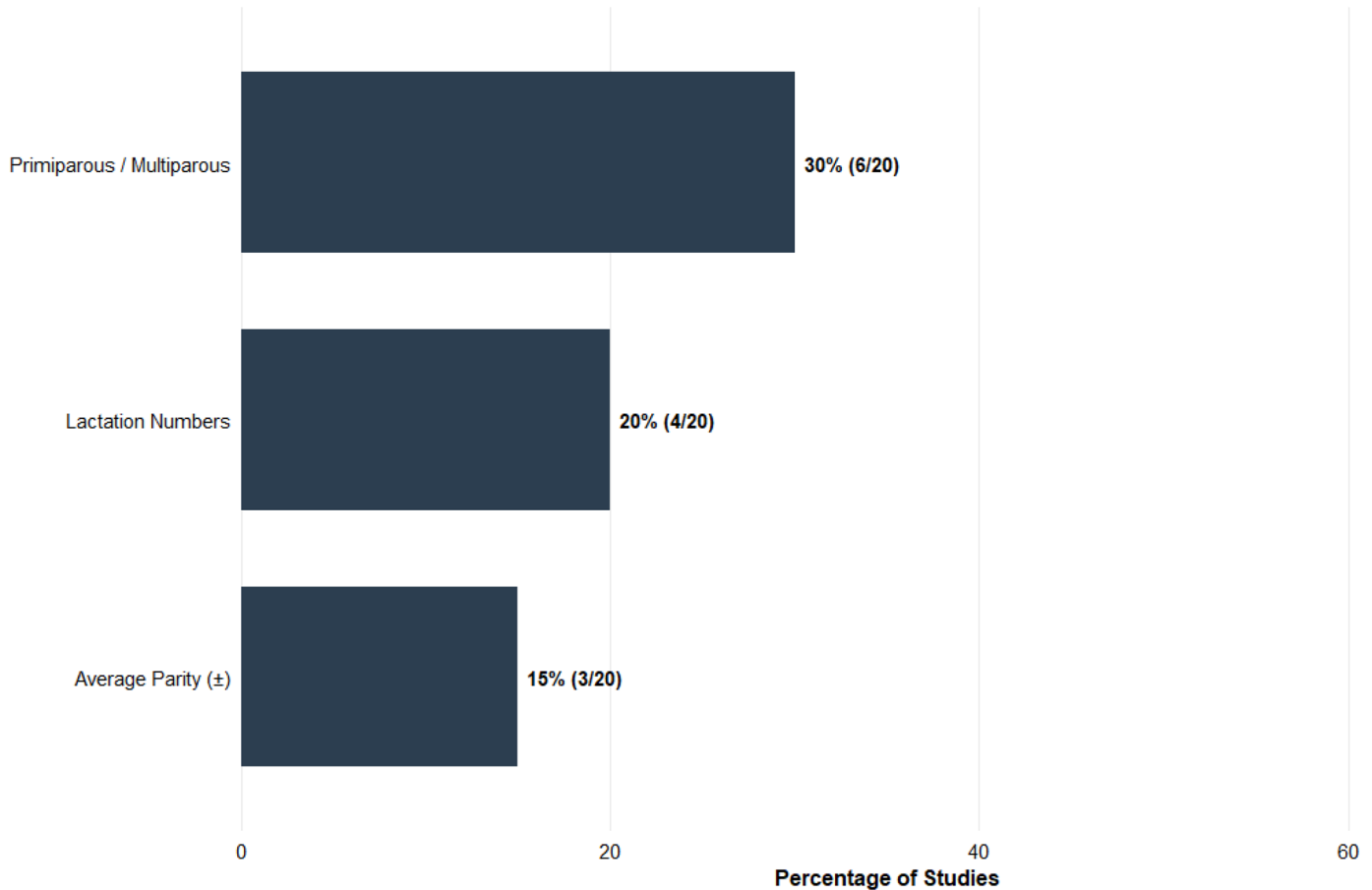


Figure 5. The percentage of studies based on how lactation was reported. Lactation reporting methods are shown on the y-axis, with the x-axis representing the percentage of included studies ($n = 20$). Categories include: Primiparous/multiparous (studies that classified cows by parity status), lactation number (studies that reported discrete lactation numbers), and average parity (studies that reported the mean and SD of parity). Bars are labeled to the right with the percentage and proportion of studies reporting each method.

assessments. Likewise, Mineur et al. (2017) reported that “lameness was assessed by trained technicians” without specifying how many technicians were involved.

Second, several studies introduced ambiguity about whether a single person conducted all assessments. For example, Anagnostopoulos et al. (2021) reported, “All feet were lifted and examined by a qualified veterinarian,” without clarifying if the same veterinarian conducted all assessments throughout the study. Coşkun et al. (2023) similarly stated that scoring was performed “by an assessor in June 2021,” leaving ambiguity about whether the same assessor evaluated all cows. Other examples include Ghotoorlar et al. (2012), who noted that cows were gait scored “by an experienced observer,” and Aravamuthan et al. (2024a,b), who reported that images were scored “by a trained investigator,” but none of the studies clarified whether a single evaluation conducted all assessments.

Third, some studies provided no information about the number or identity of the evaluators. For instance, He et al. (2022) stated that lame and healthy cows were identified based on visual assessment using the Agriculture and Horticulture Development Board (AHDB) scoring system (AHDB, 2023) but did not specify who performed the scoring. Similarly, Altay and Delialioğlu (2022) described locomotion scoring based on Sprecher et al. (1997) without identifying the evaluators responsible for the assessments.

Among the studies that did report the number of evaluators, they provided detailed descriptions of the observers. For examples using 1 observer, Busin et al. (2019) noted, “The mobility score was carried out by the same observer,” and Volkmann et al. (2021) stated, “Hoof trimming on-farm was performed by an external professional hoof trimmer,” further clarifying that “affected limbs were documented by the professional hoof trimmer.” Other studies explicitly reported the use of

Table 3. The model training characteristics reported in the included studies¹

Study authors	Total sample size	Training set size	Testing set size	Cross-validation type
Altay and Delialioğlu, 2022	40	Not stated	Not stated	Not stated
Anagnostopoulos et al., 2021	12,221	CV	CV	10-fold
Aravamuthan et al., 2024a	Min: 409; Max: 2,227	90%	10%	
Aravamuthan et al., 2024b	530	90%	10%	
Barden et al., 2021	Min: 58; Max: 211	CV	CV	5-fold
Busin et al., 2019	51	CV	CV	LOOCV
Coşkun et al., 2023	33	CV	CV	5-fold
He et al., 2022	21	CV	CV	LOOCV
Kang et al., 2020	456	60%	20%	
Kang et al., 2022	1,500	66%	33%	
Lavrova et al., 2023	20,247 Measurements then removed measures	80%	20%	
Lemmens et al., 2023	Min: Not stated; Max: 574	80%	20%	
Mineur et al., 2017	Min: 325; Max: 10,606	66%	33%	
Ghotoorlar et al., 2012	105	60%	40%	
Neupane et al., 2024	Min: 31; Max: Not stated	CV	CV	5-fold
Pastell and Kujala, 2007	73	50%	50%	
Post et al., 2021	48,041 from 3 farms	CV train/test	CV train/test	5-fold (same farm)
Volkman et al., 2021	640	75%	25%	
Zhao et al., 2023	212	CV	CV	10-fold
Zhou et al., 2023	Herd 1: Not stated; Herd 2: 2,437	Herd 1: 1,618	Herd 1: Not stated; Herd 2: 819	

¹This table summarizes the total sample size, training set size, testing set size, and type of cross validation used, as reported by each study.

multiple observers. Kang et al. (2020) stated, “lameness was manually scored by two trained observers,” and Kang et al. (2022) similarly reported, “lameness was manually scored by two experts for validation.” Neupane et al. (2024) involved 2 evaluators, where one study personnel collected locomotion scores and one

hoof trimmer recorded hoof lesions. Zhou et al. (2023) used 3 evaluators, with the final score determined by majority agreement or, when all scores differed, by the median value. Barden et al. (2021) provided a clear example of transparent reporting involving 4 evaluators, noting that over 95% of assessments were performed by

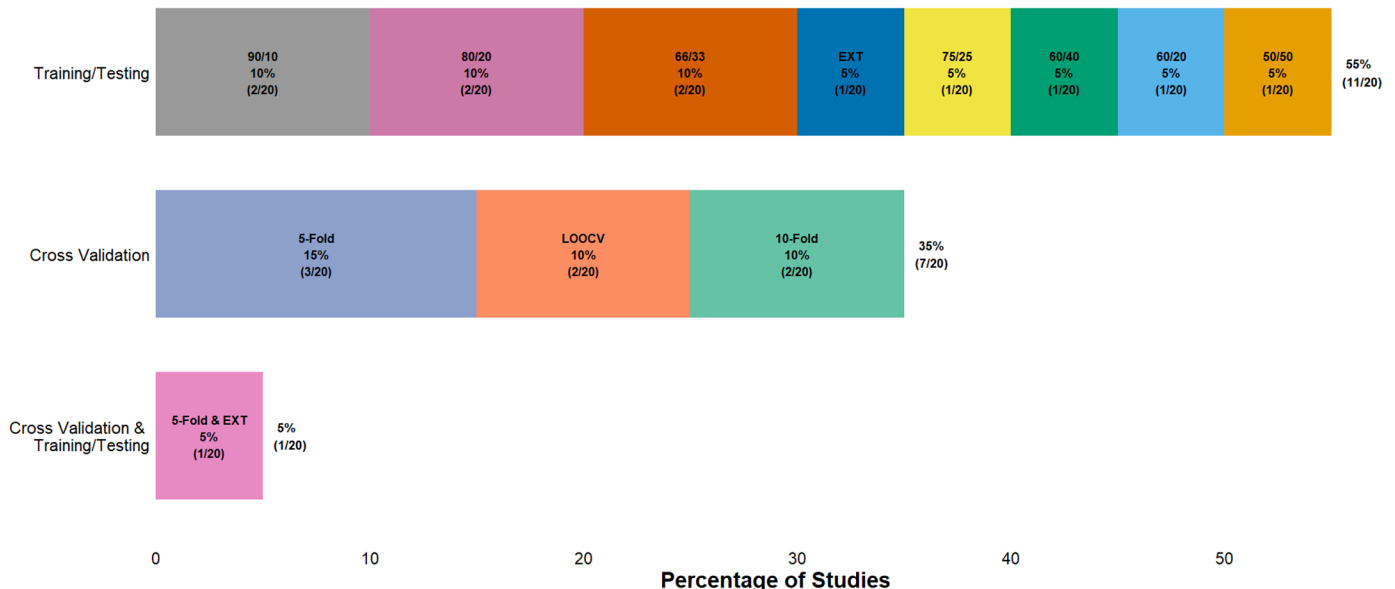


Figure 6. The proportion of studies by validation approach and implementation method (n = 20). The y-axis categorizes the type of validation used: testing/training, cross validation, or a combination of both. The x-axis shows the percentage of studies using each approach. For testing/training, labels indicate the data split (e.g., 80/20 for training/testing) or use of an external dataset (EXT). For cross-validation, labels specify the fold type (i.e., 5-fold, 10-fold, or leave-one-out cross validation [LOOCV]). For combined methods, both validation types were applied. Each segment is annotated with the percentage and number of studies using that specific approach. The total percentage and count of studies using each main validation type are displayed to the right of each full bar.

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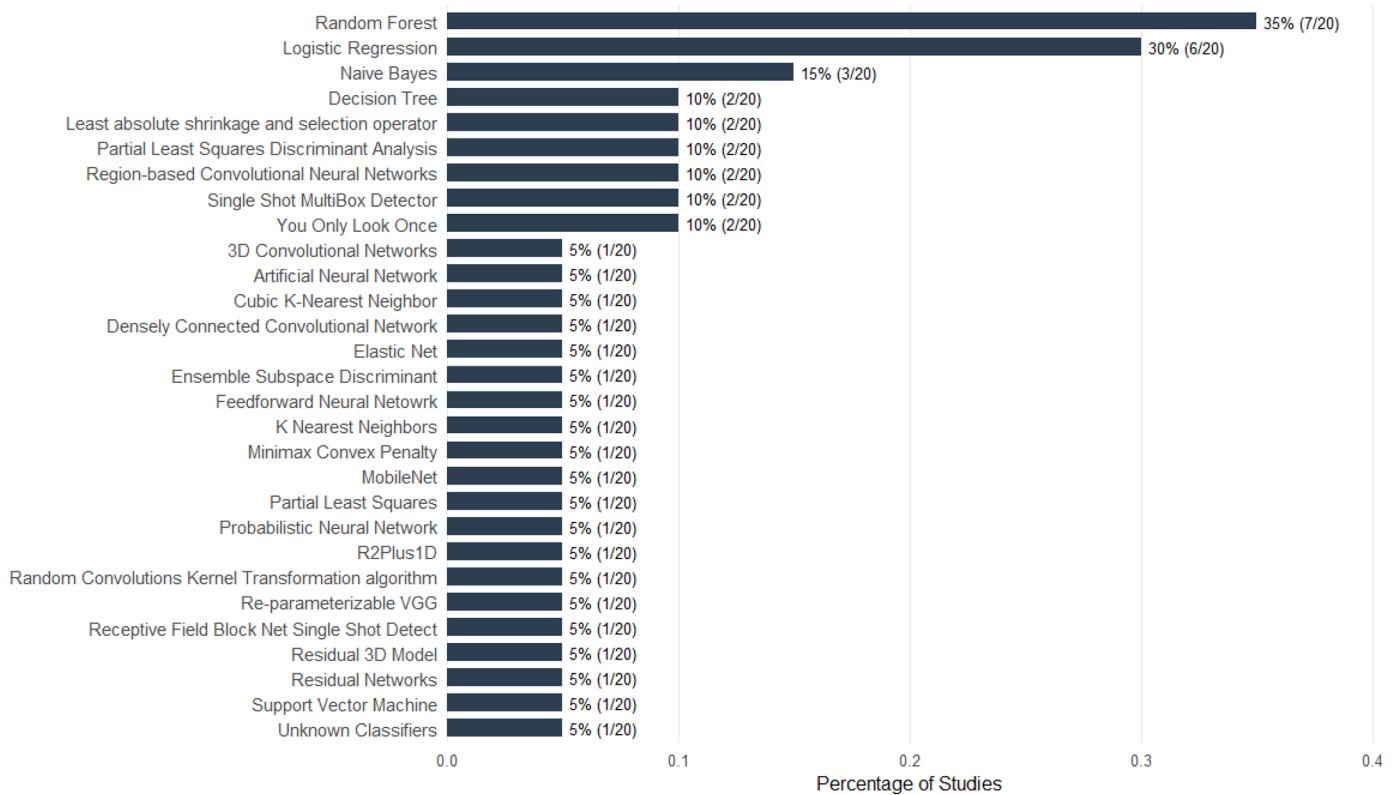


Figure 7. The percentage of studies by machine learning algorithm used. The y-axis lists the machine learning algorithms applied across included studies (n = 20), and the x-axis represents the percentage of studies that used each algorithm. Each bar is labeled to the right with the percentage and the number of studies using the algorithm out of the total.

a single researcher, while the remainder were completed by 3 additional researchers.

The lack of details around evaluators limits our ability to replicate study conditions and understand how evaluator-related variation may have influenced model outcomes. Regardless of how evaluators are described or how many are involved, consistent adherence to standardized outcome definitions is essential to ensure data reliability. Without consistent application of outcome definitions, machine learning models risk being trained on inconsistently labeled data, reducing both robustness and reproducibility. Together, these issues highlight the role of standardized outcome definitions and the development of generalizable machine learning algorithms.

Study Outcomes

Defining and classifying outcomes is a critical step in machine learning model development, as the intended use for an algorithm is dependent on how outcomes are defined. For example, a model designed to identify cows with impaired locomotion must be trained on locomotion scores, while a model aimed at creating hoof trimming lists may incorporate a variety of hoof health indicators

as its outcome. Similarly, models built to monitor hoof lesion severity must be developed to differentiate among stages of a hoof lesion. In each case, the outcome definitions guide the model's development and determine its use on-farm.

In this review, we found that studies varied considerably in how they defined, referenced, and applied outcome criteria for locomotion scores and hoof lesions (Table 5). Locomotion scores were the most commonly reported outcome, followed by hoof lesions, and then outcomes that incorporated both locomotion scores and hoof lesion data (Figure 11). Of the studies that used both locomotion scores and hoof lesions, Neupane et al. (2024) reported both locomotion scores and hoof lesions as 2 separate outcomes, while Pastell and Kujala (2007) combined locomotion scores and hoof lesion observations to create a single outcome.

Locomotion Score Outcomes. Among the studies reviewed, a small proportion of studies (16.67% or 2/12) used a 3-point scale, citing both Engel et al. (2003) and O'Callaghan et al. (2003; Figure 12). However, Engel et al. (2003) referenced the earlier 9-point system by Manson and Leaver (1988), and O'Callaghan et al. (2003) described a 5-point scale. Similarly, 16.67% (2/12) of

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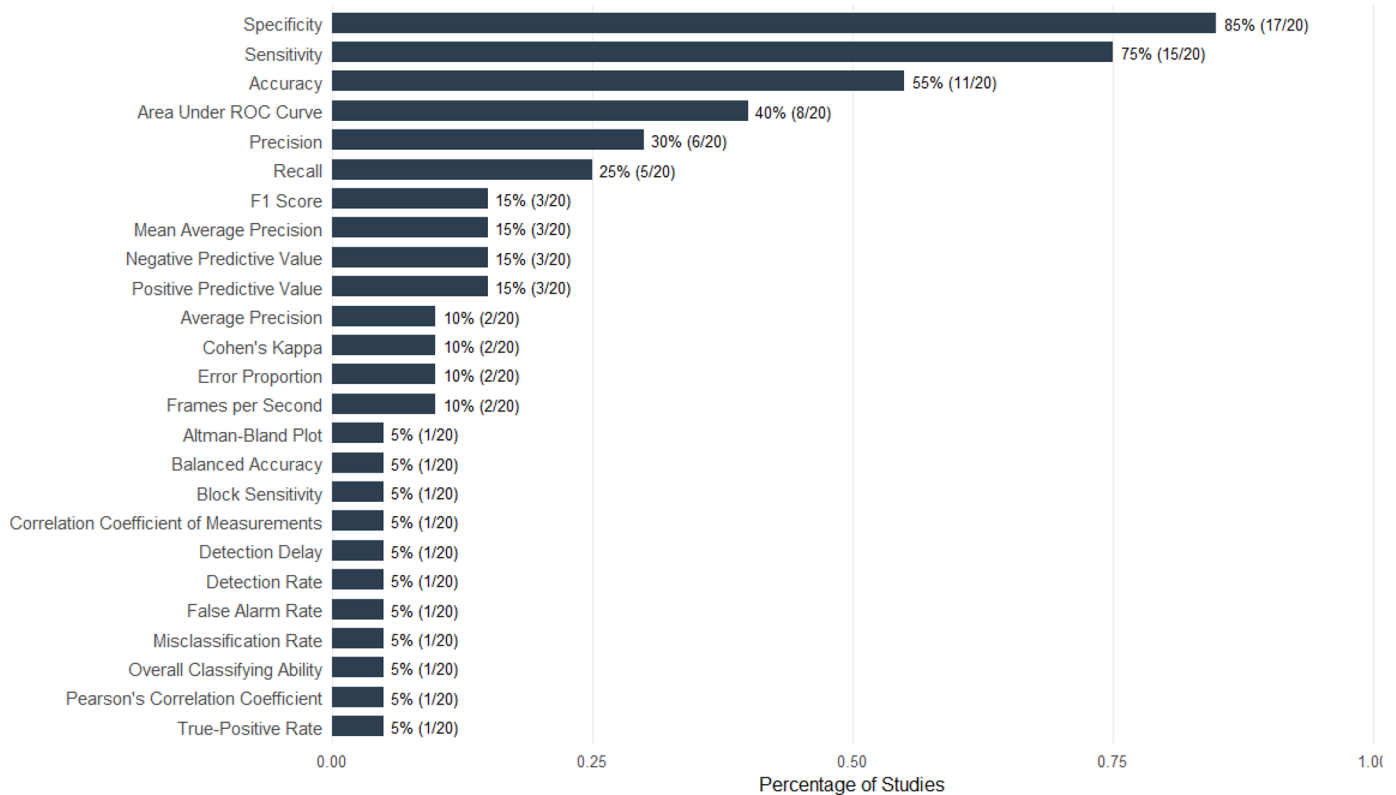


Figure 8. The percentage of studies by evaluation metric used to assess model performance. The y-axis lists the evaluation metrics reported across included studies (n = 20), while the x-axis represents the percentage of studies using each metric. Each bar is labeled to the right with the percentage and the number of studies that used that metric. ROC = receiver operating characteristic.

studies used a 4-point scale, referencing either Whay et al. (2003) or the earlier mentioned AHDB scoring system. The majority of studies (66.67% or 8/12) used a 5-point

locomotion scoring scale, most often citing Sprecher et al. (1997), with few referencing the 5-point locomotion scoring scale by Flower and Weary (2006).

Table 4. The outcome evaluators and number of evaluators used in each study¹

Study	Outcome evaluator(s)	Number of evaluators
Altay and Delialioğlu, 2022	Not stated	Not stated
Anagnostopoulos et al., 2021	Veterinarian(s)	Unclear if 1 or more
Aravamuthan et al., 2024a	Trained investigator(s)	Unclear if 1 or more
Aravamuthan et al., 2024b	Trained investigator(s)	Unclear if 1 or more
Barden et al., 2021	Veterinarian(s)	4
Busin et al., 2019	Veterinarian(s)	1
Coşkun et al., 2023	An assessor	Unclear if 1 or more
He et al., 2022	Not stated	Not stated
Kang et al., 2020	Trained observer(s)	2
Kang et al., 2022	Experts	2
Lavrova et al., 2023	Veterinarian(s)	Multiple but unclear
Lemmens et al., 2023	Employees of company and members from research group	Multiple but unclear
Mineur et al., 2017	Trained technicians(s)	Multiple but unclear
Ghotoorlar et al., 2012	Experienced observer(s)	Unclear if 1 or more
Neupane et al., 2024	Study personnel and hoof trimmer(s)	2
Pastell and Kujala, 2007	Experienced personnel and veterinarian(s)	Multiple but unclear
Post et al., 2021	Veterinarian(s)	Unclear if 1 or more
Volkman et al., 2021	Hoof trimmer(s)	1
Zhao et al., 2023	Experienced observer(s)	3
Zhou et al., 2023	Hoof trimmer(s) and veterinarian(s)	Multiple but unclear

¹This table summarizes who was responsible for outcome classification in each study, along with the number of evaluators involved in assessing the outcome.

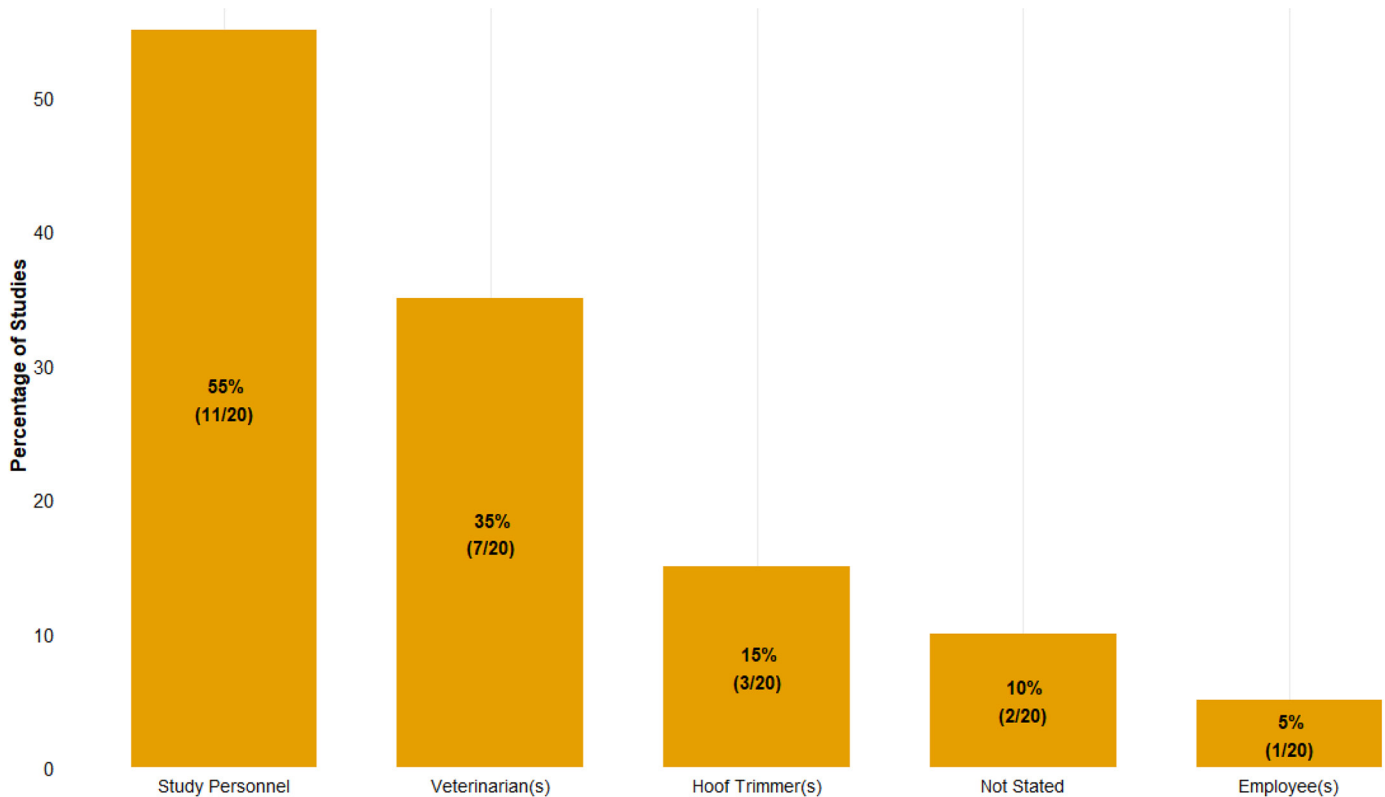


Figure 9. The percentage of studies by classification of outcome evaluator. The x-axis represents evaluator categories, and the y-axis shows the percentage of included studies ($n = 20$) that identified each type of evaluator. Evaluators labeled as “Expert,” “Assessor,” or identified with general terms such as “Observer,” “Investigator,” “Research Group,” “Technician,” or “Personnel” were grouped under study personnel. References to “Veterinarian” were categorized as veterinarian(s), mentions of “Hoof Trimmer” as hoof trimmer(s), and any use of “Employee” as employee(s). Studies that did not specify the evaluator were grouped as not stated. Each bar is labeled with the percentage and the number of studies using each evaluator classification.

Although most studies referenced established scales, the scale assessment criteria varied in detail provided (Supplemental Table S1, see Notes). While studies referenced established scales such as those developed by Whay et al. (2003) and O’Callaghan et al. (2003), these sources provided general score descriptions from 0 (sound) to 3 (severely lame) but did not include detailed assessment criteria, limiting clarity on how locomotion scores were defined. The most commonly cited reference, Sprecher et al. (1997), did provide assessment criteria; however, when addressing lameness scores of 1 (normal) and 2 (mildly lame), the criteria lacked specificity by failing to define what constitutes a normal gait. In contrast, more recent scales, such as the AHDB (2023) and Flower and Weary (2006), offered clearer and more descriptive assessment criteria for each locomotion score.

These inconsistencies in locomotion scale choice and assessment criteria highlight a challenge in establishing a commonly referenced scale across the dairy industry. Mason et al. (2022) argue that global agreement on a single locomotion scale is unlikely due to the diversity of production systems worldwide. Nonetheless, it is es-

sential that studies clearly describe the criteria used for locomotion score classification, regardless of the scale applied. This includes defining the levels in locomotion scoring scale as well as providing clear, repeatable assessment criteria for what constitutes a normal gait. In addition, studies should also describe how deviations from the normal gait are identified in a way that minimizes self-interpretation, thereby reducing subjectivity.”

Beyond defining individual locomotion scores, studies should report how locomotion scores are interpreted and whether the outcomes reflect condensed versions of the original scale (e.g., lame vs. not lame). Any decision to collapse a detailed locomotion scale should be justified, especially if a pre-existing scale already reflects those groupings. Additionally, if scores are grouped, then the grouping criteria used to form these categories should be described.

In addition to the locomotion score scales varying across studies, the grouping of locomotion scores to create outcome definitions also varied (Figure 13). Several studies simplified locomotion scoring into a binary outcome, categorizing cows as either lame or not lame.

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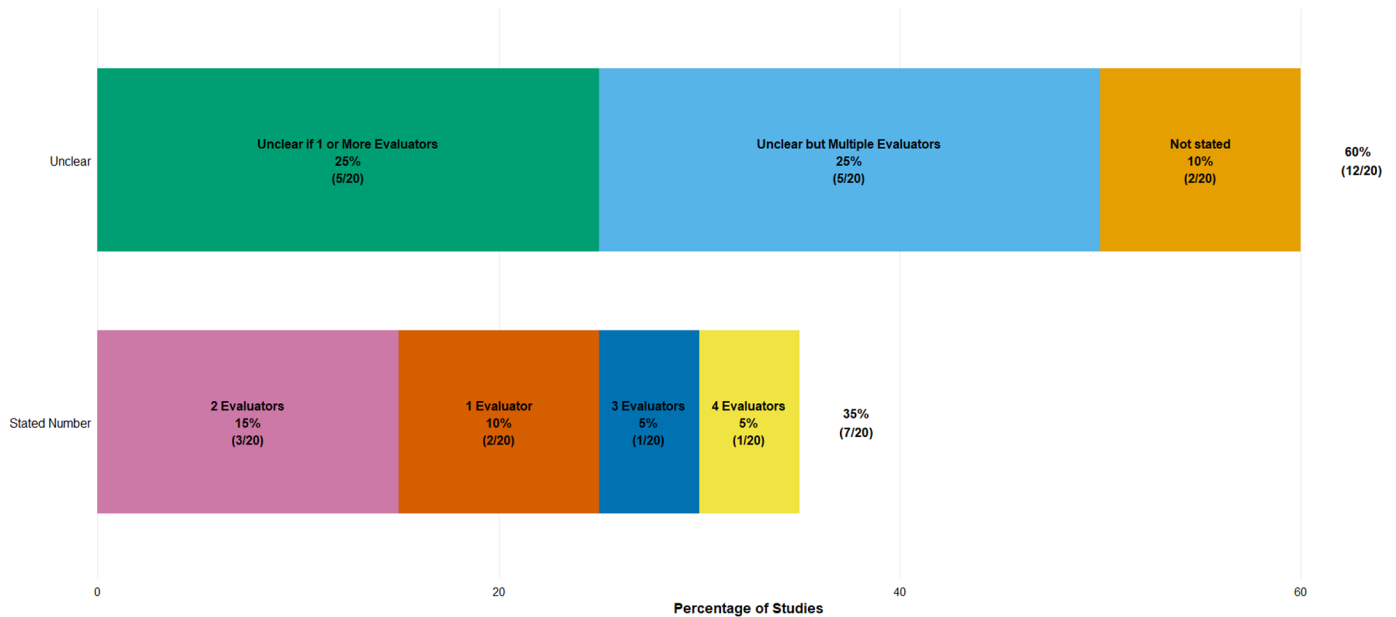


Figure 10. The percentage of studies by clarity in reporting the number of outcome evaluators ($n = 20$). The x-axis shows the percentage of studies, while the y-axis groups studies into 2 categories: those that clearly stated the number of evaluators (stated number) and those where the number of evaluators was unclear or not reported (unclear). Within the stated number category, bars indicate how many evaluators were reported (i.e., 1, 2, 3, or 4). The unclear category includes studies that either mentioned evaluators without specifying a number or did not state the number of evaluators. Specifically, these include unclear if 1 or more evaluator, unclear but multiple evaluators, and not stated. Each bar segment is labeled with the percentage and proportion of studies. The total percentage and count for each category are shown at the end of each bar.

Table 5. The outcome type reported by each study hoof lesion (HL) or locomotion score (LS)¹

Study	Outcome types	Outcome reference
Altay and Delialioğlu, 2022	LS: 5-Point scale	Sprecher et al. (1997)
Anagnostopoulos et al., 2021	HL: Digital dermatitis	Berry et al. (2012)
Aravamuthan et al., 2024a	HL: Digital dermatitis	Döpfer et al. (1997); Berry et al. (2012); Cernek et al. (2020); Anklam et al., 2022(a; b)
Aravamuthan et al., 2024b	HL: Digital dermatitis	Döpfer et al. (1997); Berry et al. (2012)
Barden et al., 2021	HL: Sole hemorrhage, sole ulcer	Egger-Danner et al. (2020)
Busin et al., 2019	LS: 4-Point scale	Whay et al. (2003)
Coşkun et al., 2023	LS: 5-Point scale	Sprecher et al. (1997)
He et al., 2022	LS: 4-Point scale	AHDB (2023)
Kang et al., 2020	LS: 3-Point scale	Engel et al. (2003); O'Callaghan et al. (2003)
Kang et al., 2022	LS: 3-Point scale	Engel et al. (2003); O'Callaghan et al. (2003)
Lavrova et al., 2023	LS: 5-Point scale	Sprecher et al. (1997)
Lemmens et al., 2023	LS: 5-Point scale	Sprecher et al. (1997)
Mineur et al., 2017	LS: 5-Point scale	Sprecher et al. (1997)
Ghotoorlar et al., 2012	LS: 5-Point scale	Sprecher et al. (1997)
Neupane et al., 2024	LS: 5-Point scale; HL: Digital dermatitis foot rot injury	LS: Sprecher et al. (1997); HL: Not stated
Pastell and Kujala, 2007	TLS (undefined) white line disease	
	LS: 5-Point scale; HL: Hemorrhages interdigital phlegmon sole ulcers white line problems	LS: Sprecher et al. (1997); HL: Not stated
Post et al., 2021	HL: Lameness treatment	Not stated
Volkman et al., 2021	HL: Digital dermatitis, double sole, heel horn erosion, horn fissure, interdigital hyperplasia, sole hemorrhage, sole ulcer (Rusterholz)/toe ulcer, white line disease/white line abscess	AID-Infodienst (2014)
Zhao et al., 2023	LS: 5-Point scale	Flower and Weary (2006)
Zhou et al., 2023	HL: Abscesses, digital dermatitis, interdigital dermatitis, sole ulcers	Not stated

¹For HL outcomes, the specific lesion type assessed is provided; for LS outcomes, the locomotion scoring scale used is noted. The final column includes the reference(s) cited by the study to define the hoof lesion classifications or the locomotion scoring system.

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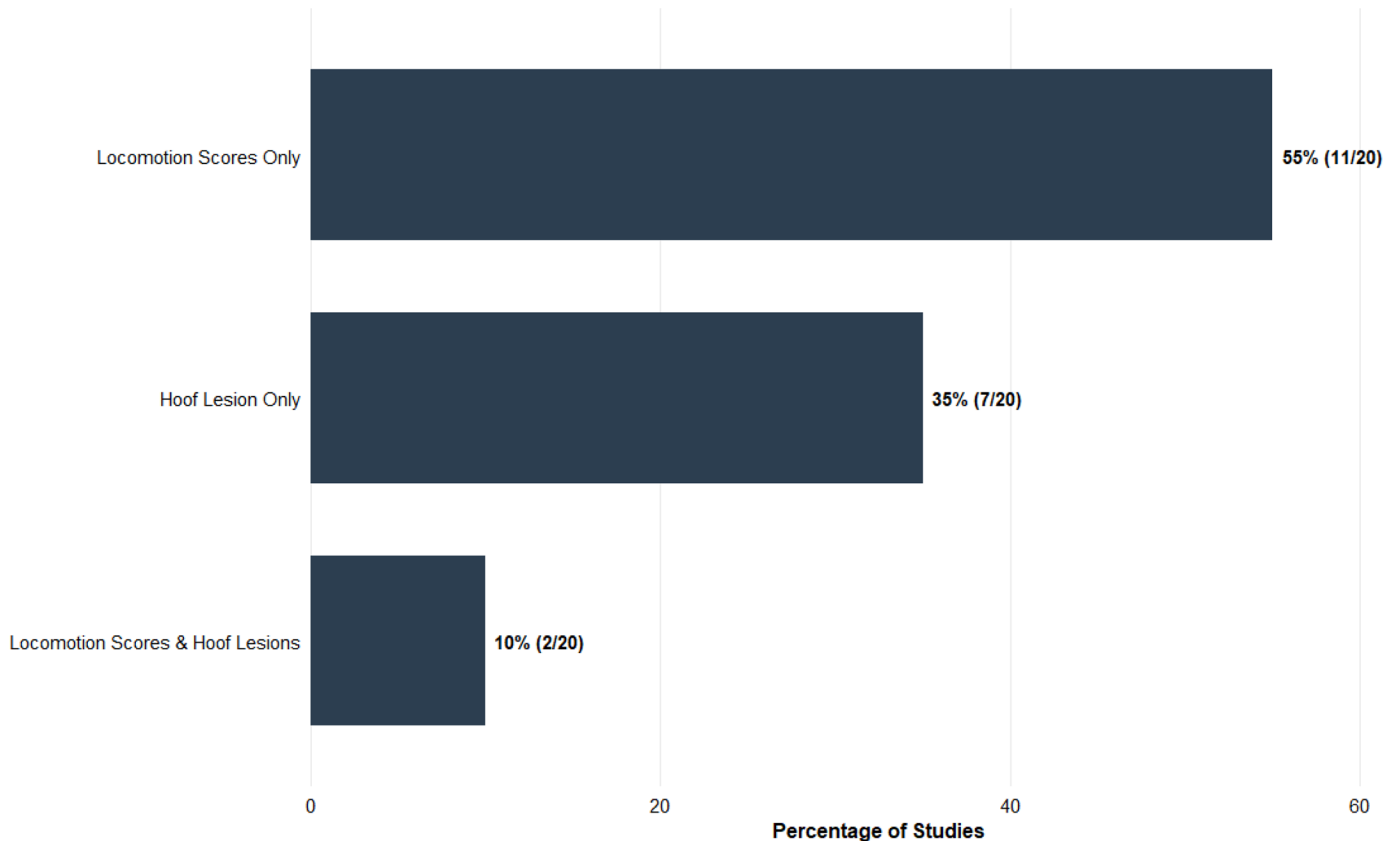


Figure 11. The percentage of studies by outcome type used for model development. The y-axis lists the outcome types used in the included studies (n = 20): locomotion scores only (studies that developed outcomes based solely on locomotion scoring), hoof lesions only (studies that developed outcomes based solely on hoof lesion), and locomotion scores and hoof lesions (studies that developed outcomes based on both). The x-axis represents the percentage of studies, and each bar is labeled to the right with the percentage and the number of studies using that outcome type.

The approach used by Busin et al. (2019) categorized cows as lame if they received a score of 1, 2, or 3 on a 4-point scale, effectively flagging any deviation from sound locomotion, score 0, as lameness. This strategy may be appropriate in contexts prioritizing early detection or heightened sensitivity, where even mild gait abnormalities are of concern. In contrast, studies applying the Sprecher et al. (1997) 5-point scale (Mineur et al., 2017; Altay and Delialioğlu, 2022; Coşkun et al., 2023; Lavrova et al., 2023) grouped cows with scores of 1 and 2 as healthy and those with scores of 3, 4, or 5 as lame. While this threshold is commonly used by studies assessing prevalence (von Keyserlingk et al., 2012; Cook et al., 2016), it is important to note that a score of 2 on the Sprecher et al. (1997) scale indicates mild lameness, meaning that mildly lame cows were grouped as healthy in these studies. With this in mind, studies using binary classification should consider whether cows with mild lameness or imperfect locomotion should be grouped with sound cows, as this decision can influence the interpretation and application of model outputs.

In contrast to the previous aim to distinguish lame cows from nonlame cows, Neupane et al. (2024) applied a binary-like classification based on severity, grouping cows with scores of 2 and 3 as subclinically lame and scores of 4 and 5 as severely lame. Although Neupane et al. (2024) stated that separating moderately and severely lame cows may help producers prioritize management decisions, their model did not assess whether sound and lame cows could be distinguished, which limits its usefulness for on-farm detection and intervention. Without consistency in how binary outcomes are defined, it becomes difficult to compare findings or apply models across different herds or management systems that may have different detection goals.

Several studies used 3-level classifications. Kang et al. (2020, 2022) used a 3-point locomotion scoring scale and classified cows into 3 categories: no lameness, mild lameness, and severe lameness. Similarly, Zhao et al. (2023) applied a 3-level classification using the 5-point Flower and Weary (2006) scale, grouping cows with scores of 1 and 2 as healthy or relatively healthy, scores

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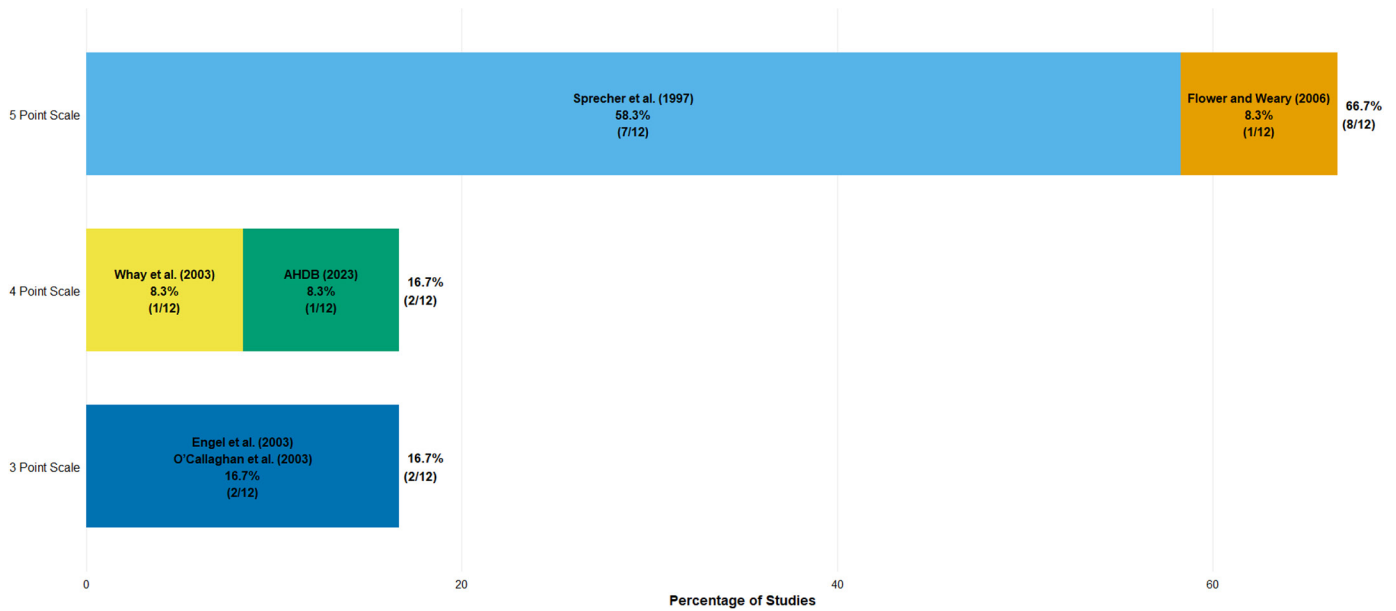


Figure 12. The percentage of studies using locomotion scores only (n = 12), categorized by locomotion score point scale and referenced scoring system. The y-axis lists the locomotion score point scales applied in the included studies. The x-axis represents the percentage of studies using each scale. Each bar is segmented by the specific scoring system referenced, with segment labels showing the percentage and proportion of studies using that reference. To the right of each full bar is the total percentage and proportion of studies using that point scale. AHDB (2023) refers to the Agriculture and Horticulture Development Board published in 2023.

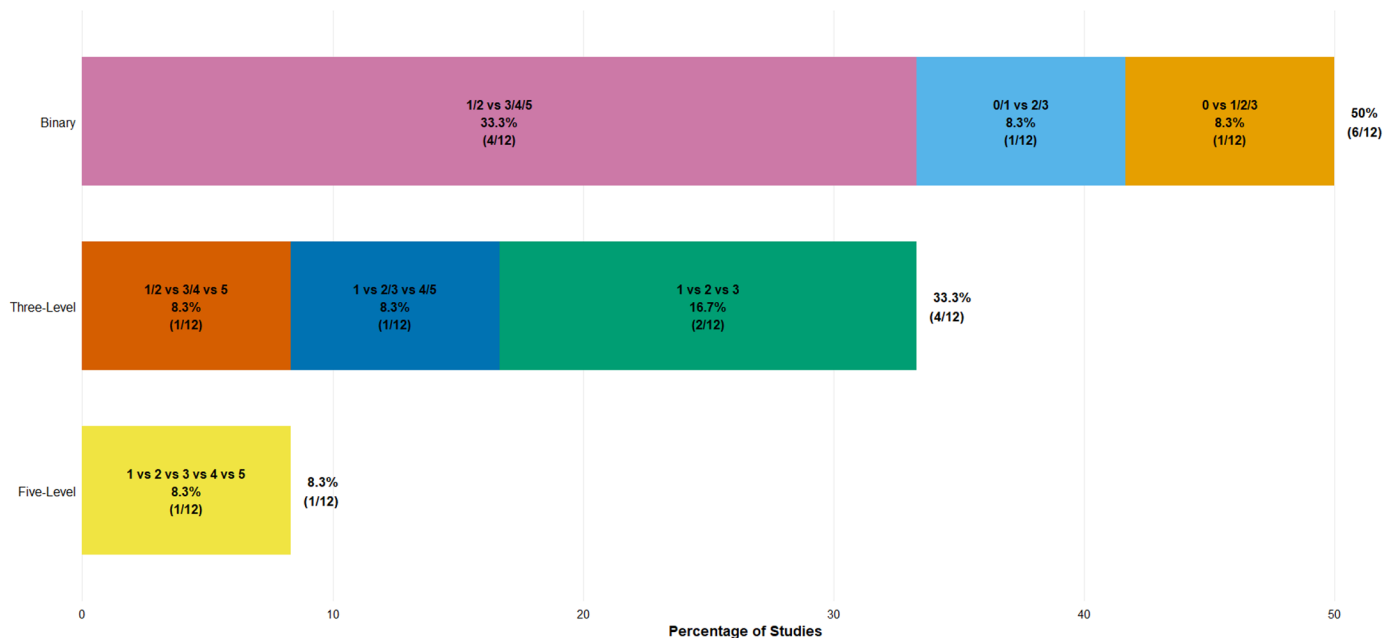


Figure 13. The percentage of studies using locomotion scores as outcomes (n = 12), categorized by the number of outcome levels. The y-axis indicates how locomotion scores were grouped to define outcome variables: Binary (2-level), 3-level, or 5-level classifications. The x-axis represents the percentage of studies using each outcome structure. Within each category, bars are segmented by how the locomotion scores were split. Each segment is labeled with the percentage and number of studies using that specific categorization. Total percentages and counts for each outcome structure are shown at the end of each bar.

of 3 and 4 as early lameness, and a score of 5 as severe lameness. In contrast, Lemmens et al. (2023) applied a more conservative classification threshold. Their 3-level system categorized cows with a score of 1 as nonlame, scores of 2 and 3 as mildly lame, and scores of 4 and 5 as severely lame. This grouping is more conservative than those used by Kang et al. (2020, 2022) and Zhao et al. (2023), as it separates cows with any mobility impairment from sound cows. This approach is conceptually similar to that of Busin et al. (2019), who grouped cows with any deviation from a sound locomotion score into the lame category in their binary classification.

Although these 3-level approaches vary in their grouping criteria, they align conceptually with the 3-point locomotion scoring system used by the National Dairy Farmers Assuring Responsible Management (FARM) Program in the United States, which classifies cows as normal, moderate, or severely lame based on their mobility (FARM Animal Care Version 5). Given this alignment, automated 3-level classification algorithms could potentially replace the need for human observers in welfare assessments, such as the FARM locomotion scoring programs, which focus on identifying severely lame cows. A precedent for this approach already exists in the United Kingdom, where mobility scores generated by a commercial automated camera technology are accepted by UK Arla milk suppliers as an alternative to quarterly mobility scores collected by trained human observers (CattleEye, 2023). This example demonstrates how the previously described 3-level approaches could be used in both research and production settings.

While most studies condensed locomotion scores into 2 or 3 categories, only one study retained the full 5-level locomotion scoring scale without collapsing categories (Ghotoorlar et al., 2012). Preserving all 5 levels may be appropriate in research settings where detailed locomotion assessments are necessary, such as when investigating the progression and severity of lameness over time or when developing predictive models that benefit from a higher level of detail. This approach allows for more detailed data collection and analysis, which may support herd-level research or epidemiological studies. However, in commercial production environments, the practical application of a 5-level classification may be limited. For example, Whay (2002) suggests that the goal of visual locomotion scoring for producers should be to identify cows that need to be referred to a hoof trimmer for diagnosis and treatment. This type of simplified outcome supports automation and reduces labor demands, which are commonly preferred in commercial settings (Dela Rue et al., 2019).

Hoof Lesion Outcomes. Similar to locomotion scores, studies that used hoof lesions as an outcome varied in how hoof lesions were defined, referenced, and classified

(Table 5). Although some studies cited established references (described in Supplemental Table S2, see Notes), many did not provide hoof lesion descriptions or cite any sources. Even among cited references, hoof lesion definitions varied, and studies often focused on different hoof lesion types, complicating the description of hoof lesion definitions.

Among studies in this review, there were 3 (Anagnostopoulos et al., 2021; Aravamuthan et al., 2024a,b) that focused on differentiating between stages of digital dermatitis, and the sources used to describe the digital dermatitis stages overlapped (Table 5). In particular, Aravamuthan et al. (2024a,b) focused on performing a multiclass classification approach using computer vision to distinguish between each M-stage: M0, M2, M2P, M4H, and M4P, which is described in Supplemental Table S2. Briefly, M0 represents normal digital skin; M2 is an active ulcerative or granulomatous lesion greater than 20 mm in diameter; and M2P lesions are M2 lesions with additional proliferative growth. The chronic stages include M4H and M4P, which are characterized by hyperkeratosis and proliferative tissue but are not active, respectively.

In addition, Aravamuthan et al. (2024a) grouped lesions by differentiating healthy or chronic stages (M0, M4H) from active, painful stages (M2, M2P). Similarly, Anagnostopoulos et al. (2021) categorized stages based on activity, initially grouping feet as either having active lesions (M2) or without active digital dermatitis lesions. They also implemented an additional classification of active lesions by including M1, M2, and M4.1 stages. The identification of active lesions is clinically meaningful, as these lesions are identified for treatment. While all studies included the M2 stage in their definition of active lesions, there was variation in whether M1 and M4.1 were also included. This inconsistency is not unexpected, as reviewed by Cook (2017), who noted that there are differing perspectives on how “active” lesions should be defined. Some studies consider only M2 as active, whereas others include M1 and M4.1 as well. Regardless of differences in treatment recommendations for M1 and M4.1, their detection may still provide valuable insight into disease dynamics and support herd-level surveillance and early outbreak detection (Tremblay et al., 2016).

In contrast, among the studies that incorporated more hoof lesions than just digital dermatitis, only Volkmann et al. (2021) and Barden et al. (2021) described hoof lesions and referenced sources, AID-Infodienst (2014) and the ICAR Claw Health Atlas (Egger-Danner et al., 2020), respectively. Though their outcomes varied, Barden et al. (2021) focused on differentiating cows with sole hemorrhages or sole ulcers from those without, while Volkmann et al. (2021) categorized cows as “healthy” or “affected” based on the presence of at least one lesion,

including digital dermatitis, double sole, heel horn erosion, horn fissure, interdigital hyperplasia, sole hemorrhage, sole ulcer (Rusterholz/toe ulcer), and white line disease/abscess. Despite these differences, both studies included overlapping hoof lesions of sole hemorrhages and sole ulcers.

AID-Infodienst (2014) and ICAR Claw Health Atlas provided nearly identical descriptions of sole hemorrhages, defining them as “diffused or circumscribed red or yellow discoloration of the sole or white line” (Supplemental Table S2). Both sources also described sole ulcers, though ICAR Claw Health Atlas offered additional detail, specifying that fresh or necrotic corium must be visibly exposed. Even though Volkmann et al. (2021) had included additional lesion types in their outcome and ICAR Claw Health Atlas provides definitions for many of those lesions, Barden et al. (2021) did not include them as outcomes. Therefore, it is not appropriate to compare hoof lesion definitions for outcomes that were not used in both Volkmann et al. (2021) and Barden et al. (2021).

In contrast to Volkmann et al. (2021) and Barden et al. (2021), other studies included hoof lesions in their outcomes but did not provide definitions. Neupane et al. (2024) grouped digital dermatitis and foot rot as infectious lesions, while white line disease, TLS (not defined), and injury were classified as noninfectious, with cows categorized as having either infectious or noninfectious lesions. Zhou et al. (2023) differentiated cows based on the absence/presence of at least one lesion (abscesses, digital dermatitis, interdigital dermatitis, sole ulcers).

Neither of these studies included detailed descriptions of hoof lesions, despite the availability of standardized definitions in resources such as the aforementioned AID-Infodienst (2014) and the ICAR Claw Health Atlas. For example, the ICAR Claw Health Atlas considers interdigital phlegmon and foot rot to be synonymous terms and defines interdigital dermatitis as mild dermatitis around the claws that is not classified as digital dermatitis. The ICAR Atlas only refers to abscesses in the context of white line abscesses. The abbreviation TLS was used in one study without clarification. The lack of standardized definitions and absence of lesion descriptions in these studies limits the ability to interpret how hoof lesions were classified and if they align across studies, reducing the comparability of findings across studies.

Although Post et al. (2021) defined outcomes based on treatment history and did not provide descriptions of what the lameness treatments were, they did use lameness treatments as a proxy for hoof lesion presence and accounted for whether cows had previously received treatment, making it one of the few studies to account for lesion reoccurrence. Notably, Post et al. (2021) showed that including treatment history improved model perfor-

mance, highlighting the importance of longitudinal lesion information. Similarly, Barden et al. (2021) was the only study in this review to explicitly categorize outcome definitions based on both lesion reoccurrence and severity. They accounted for hemorrhage severity (Supplemental Table S3, see Notes) by grouping cows with severe sole hemorrhages or any sole ulcer as having active lesions and those without as control cows. Additionally, Barden et al. (2021) incorporated lesion reoccurrence in their outcome classification by differentiating between new sole hemorrhages (New SH), new sole ulcers (New SU), the presence of either a new sole hemorrhage or new sole ulcer (New SH/SU), and the presence of either a new or chronic sole ulcer (All SU). Though the hoof lesion definitions and accompanying descriptions may be lacking, identifying cows with hoof lesions could support management decisions by prompting interventions, such as the development of hoof trimming lists.

Locomotion Score and Hoof Lesion Outcomes. While the previously described studies developed machine learning algorithms to detect either hoof lesions or locomotion scores, only Pastell and Kujala (2007) combined both measures into a single outcome. In their study, they incorporated the Sprecher et al. (1997) 5-point scale alongside hoof lesion status to classify cows. Specifically, cows with a gait score of 1 or 2 were considered sound, but only if a score of 2 was not associated with a “clinical problem.” Cows that received a score of 2 accompanied by a clinical problem, or a score of 3 or higher, were classified as lame. Pastell and Kujala (2007) also evaluated the presence and severity of specific hoof lesions, including hemorrhages, white line problems, sole ulcers, and interdigital phlegmon. Although they reported lesion severity grades (Supplemental Table S3), they did not provide clear definitions for each lesion type. Additionally, despite incorporating lesion information into their classification system, it remained unclear how they defined the term “clinical problem” when determining outcome status. Specifically, it was not specified whether the presence of any lesion, any severity level, or only severe lesions was considered a clinical problem.

Evaluating the Impact of Outcome Classification. Despite the variability in outcome types, definitions, and grouping methods across studies, none of the reviewed studies evaluated whether animals identified by the models experienced improvement following detection or intervention. While evaluating this would require external validation, as discussed earlier in the review, the practical benefit of classifying animals as lame or having hoof lesions remains untested within the included studies.

Although detection can help identify animals in need of intervention, evidence suggests that the timing and chronicity of the condition may influence how effective that intervention will be. For example, Thomas et al.

(2015) defined acute lameness as 2 consecutive nonlame scores followed by a lame score, whereas Thomas et al. (2016) defined chronic lameness as at least 2 severely lame scores within a 6-wk period based on biweekly assessments. When cows in both categories presented with hoof lesions, those with chronic lameness had lower recovery rates compared with cows with acute lameness. These findings suggest that even when severely lame cows with hoof lesions are successfully identified and treated, the applied intervention may not always lead to meaningful improvement.

Thus, classification performance alone is not sufficient to justify on-farm adoption. Studies should evaluate whether appropriate interventions exist for the condition underlying a positive classification, such as treating hoof lesions or addressing the root cause of lameness. It is equally important to assess whether those interventions lead to measurable improvements in animal outcomes. For example, positive classifications for conditions that are not directly treatable may lead to additional labor and diagnostic costs without improving recovery. Hock and knee lesions, for instance, often require environmental changes, such as removal of the causative surface or flooring material, for recovery to occur (Cramer et al., 2024). However, identifying nonrecoverable conditions is not always undesirable. From a welfare perspective, there is value in detecting certain untreatable conditions, such as muscular ruptures or hip injuries, that warrant timely euthanasia (Cramer et al., 2024). Therefore, studies should describe the diagnoses and outcomes associated with positive classifications, including whether intervention was possible and what it led to. Doing so could help determine the practical relevance of machine learning applications in dairy cattle management.

Limitations

This review has several limitations that should be acknowledged. First, the search strategy was restricted to articles published in English, which may have resulted in the exclusion of relevant studies published in other languages, potentially limiting the range of perspectives on how hoof lesions and lameness are defined in machine learning contexts. Additionally, although this review summarized the outcome definitions and evaluation metrics reported by the included studies, it did not assess or compare the actual performance of the machine learning models themselves. As a result, it was not possible to evaluate how different outcome definitions may influence model performance or to provide informed recommendations on algorithm selection. Future reviews may benefit from a synthesis of model performance to further evaluate their reliability and applicability in dairy production systems.

However, our review identified several inconsistencies that may limit the interpretation, applicability, and generalizability of findings across studies. These include inconsistent descriptions of study population characteristics, which restrict understanding of the contexts and production settings to which models may apply. Unclear reporting of the number and role of evaluators further limits replicability and reduces confidence in the generalizability of results. In addition, variation in outcome definitions reduces the ability to make cross-study comparisons. Finally, limited use of external validation and the lack of evaluation following a positive classification restrict both confidence in model performance and understanding of their practical usefulness in a real-world setting.

CONCLUSIONS

As the application of machine learning in dairy cattle management continues to expand, particularly in the detection of hoof lesions and lameness, this review identified several issues that must be addressed to ensure progress. One of the most important findings from this review is the variation in how lameness and hoof lesion outcomes are defined, categorized, and applied across studies. Inconsistent use of locomotion scoring scales and varying or absent definitions for hoof lesions create major challenges for comparison, replication, and real-world applicability of the algorithms. While some variability may be expected due to differences in production systems or research contexts, the lack of clearly stated assessment criteria limits the ability to interpret model performance. These inconsistencies are concerning given that machine learning models must be built around clearly defined outcomes. Whether the goal is early detection, severity classification, or prevalence monitoring, the outcome definitions must directly reflect the intended application. Therefore, establishing clear and consistent definitions is essential to developing models that are both valid and practically useful.

Another limitation identified in this review is the lack of transparency in study design and reporting. Key details, such as the number and identity of evaluators, herd and farm characteristics were often missing or insufficiently described. These omissions reduce the generalizability and reproducibility making it difficult to assess the validity of results across different characteristics. Compounding this issue is the limited use of external validation; most studies solely on internal data splits for model evaluation, with few testing performance on independent herds. Without validation on different herds, it remains unclear whether models developed can generalize to other herds with differing management systems, breeds, or environments. For machine learning al-

gorithms to be successful on other herds, future research must describe herd and farm characteristics and perform external validation.

Finally, while performance metrics such as accuracy, sensitivity, and specificity were commonly reported, we found inconsistencies in terminology. In some cases, different terms were used for the same metrics. Consistent use of terminology and reporting standards, particularly those grounded in epidemiological or diagnostic test evaluation, would improve clarity and facilitate comparison across studies. However, even when performance metrics were reported, no study assessed whether appropriate interventions existed for the identified diagnosis or whether acting on it led to improved outcomes for animals or meaningful benefits for producers.

These findings emphasize the importance of a more standardized, transparent, and goal-oriented approach to machine learning in dairy cattle lameness detection. The field must move toward clearly describing outcome definitions, establish expectations for external validation, and adopt consistent performance reporting practices. These steps are essential to build confidence in machine learning tools and enhance their applicability across systems.

NOTES

Drew Swartz was partially funded by the USDA NIFA Food and Agricultural Science National Needs Graduate Fellowship in Dairy Production Systems, proposal number 2020-08149 and accession number 1025222, and by a University of Minnesota Data Science Initiative Graduate Assistantship and a Global Food Ventures Fellowship Program (2024-2025) with funding made available by Minnesota's Discovery, Research, and Innovation Economy (MnDRIVE) initiative. Drew Swartz was involved in funding and data acquisition, project management, study conceptualization, data collection, management, processing, and analysis, and manuscript preparation and review; Andre Nault was involved in study conceptualization and review of the manuscript; and Gerard Cramer was involved in study design, funding and data acquisition, project management, and review of the manuscript. During the preparation of this work, the author(s) used ChatGPT to generate R code and improve the written text. After using ChatGPT, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication. Supplemental material for this article is available at [URL]. Code and data are available at [URL]. No human or animal subjects were used, so this analysis did not require approval by an Institutional Animal Care and Use Committee or Institutional Review Board. The authors have not stated any conflicts of interest.

Nonstandard abbreviations used: AHDB = Agriculture and Horticulture Development Board; EXT = external dataset; FARM = National Dairy Farmers Assuring Responsible Management; HL = hoof lesion; ICAR = International Committee for Animal Recording; LOOCV = leave-one-out cross validation; LS = locomotion score; SH = sole hemorrhages; SU = sole ulcers.

REFERENCES

- Adams, A. E., J. E. Lombard, C. P. Fossler, I. N. Román-Muñiz, and C. A. Kopral. 2017. Associations between housing and management practices and the prevalence of lameness, hock lesions, and thin cows on US dairy operations. *J. Dairy Sci.* 100:2119–2136. <https://doi.org/10.3168/jds.2016-11517>.
- AHDB. 2023. Mobility Scoring: How to Score Your Cows | AHDB. Agriculture and Horticulture Development Board. Accessed Apr. 7, 2025. <https://ahdb.org.uk/knowledge-library/mobility-scoring-how-to-score-your-cows>.
- AID-Infodienst. 2014. Klauengesundheit Beim Rind. 3rd ed. AID Infodienst Ernährung, Landwirtschaft, Verbraucherschutz, Bonn, Germany.
- Altay, Y., and R. A. Delialioğlu. 2022. Diagnosing lameness with the random forest classification algorithm using thermal cameras and digital colour parameters. *Mediterranean Agricultural Sciences* 35:47–54. <https://doi.org/10.29136/mediterranean.1065527>.
- Anagnostopoulos, A., M. Barden, J. Tulloch, K. Williams, B. Griffiths, C. Bedford, M. Rudd, A. Psifidi, G. Banos, and G. Oikonomou. 2021. A study on the use of thermal imaging as a diagnostic tool for the detection of digital dermatitis in dairy cattle. *J. Dairy Sci.* 104:10194–10202. <https://doi.org/10.3168/jds.2021-20178>.
- Aravamuthan, S., P. Cernek, K. Anklam, and D. Döpfer. 2024a. Comparative analysis of computer vision algorithms for the real-time detection of digital dermatitis in dairy cows. *Prev. Vet. Med.* 229:106235. <https://doi.org/10.1016/j.prevetmed.2024.106235>.
- Aravamuthan, S., E. Walleiser, and D. Döpfer. 2024b. Benchmarking analysis of computer vision algorithms on edge devices for the real-time detection of digital dermatitis in dairy cows. *Prev. Vet. Med.* 231:106300. <https://doi.org/10.1016/j.prevetmed.2024.106300>.
- Arksey, H., and L. O'Malley. 2005. Scoping studies: Towards a methodological framework. *Int. J. Soc. Res. Methodol.* 8:19–32. <https://doi.org/10.1080/1364557032000119616>.
- Barden, M., B. Li, A. Anagnostopoulos, B. Griffiths, C. Bedford, A. Psifidi, G. Banos, and G. Oikonomou. 2021. Can we breed claw horn lesions out of dairy cows? *Cattle Pract.* 29:72.
- Blackie, N., E. C. L. Bleach, J. R. Amory, and J. R. Scaife. 2013. Associations between locomotion score and kinematic measures in dairy cows with varying hoof lesion types. *J. Dairy Sci.* 96:3564–3572. <https://doi.org/10.3168/jds.2012-5597>.
- Bleeker, S. E., H. A. Moll, E. W. Steyerberg, A. R. T. Donders, G. Derksen-Lubsen, D. E. Grobbee, and K. G. M. Moons. 2003. External validation is necessary in prediction research: A clinical example. *J. Clin. Epidemiol.* 56:826–832. [https://doi.org/10.1016/S0895-4356\(03\)00207-5](https://doi.org/10.1016/S0895-4356(03)00207-5).
- Busin, V., L. Viora, G. King, M. Tomlinson, J. Lekernec, N. Jonsson, and F. Fioranelli. 2019. Evaluation of lameness detection using radar sensing in ruminants. *Vet. Rec.* 185:572. <https://doi.org/10.1136/vr.105407>.
- CattleEye. 2023. CattleEye approved for use on Arla 360 UK Dairies. Accessed Apr. 7, 2025. <https://cattleeye.com/news/cattleeye-approved-for-use-on-arla-360-and-arla-garden-uk-dairies/>.
- Chen, W., E. White, and N. M. Holden. 2016. The effect of lameness on the environmental performance of milk production by rotational grazing. *J. Environ. Manage.* 172:143–150. <https://doi.org/10.1016/j.jenvman.2016.02.030>.

- Cook, N. B. 2017. A review of the design and management of footbaths for dairy cattle. *Vet. Clin. North Am. Food Anim. Pract.* 33:195–225. <https://doi.org/10.1016/j.cvfa.2017.02.004>.
- Cook, N. B., J. P. Hess, M. R. Foy, T. B. Bennett, and R. L. Brotzman. 2016. Management characteristics, lameness, and body injuries of dairy cattle housed in high-performance dairy herds in Wisconsin. *J. Dairy Sci.* 99:5879–5891. <https://doi.org/10.3168/jds.2016-10956>.
- Coşkun, G., Ö. Şahin, R. A. Delialioğlu, Y. Altay, and I. Aytekin. 2023. Diagnosis of lameness via data mining algorithm by using thermal camera and image processing method in Brown Swiss cows. *Trop. Anim. Health Prod.* 55:50. <https://doi.org/10.1007/s11250-023-03468-9>.
- Cramer, G., and L. Solano. 2024. Lameness Originating in the Hoof in Cattle. Accessed Jan. 13, 2025. <https://www.merckvetmanual.com/musculoskeletal-system/lameness-in-cattle/lameness-originating-in-the-hoof-in-cattle>.
- Cramer, G., L. Solano, and P. Constable. 2024. Upper Limb Lameness in Cattle. Accessed May 13, 2025. <https://www.merckvetmanual.com/musculoskeletal-system/lameness-in-cattle/upper-limb-lameness-in-cattle>.
- Dallago, G. M., K. M. Wade, R. I. Cue, J. T. McClure, R. Lacroix, D. Pellerin, and E. Vasseur. 2021. Keeping dairy cows for longer: A critical literature review on dairy cow longevity in high milk-producing countries. *Animals (Basel)* 11:808. <https://doi.org/10.3390/ani11030808>.
- Dolecheck, K. A., M. W. Overton, T. B. Mark, and J. M. Bewley. 2019. Use of a stochastic simulation model to estimate the cost per case of digital dermatitis, sole ulcer, and white line disease by parity group and incidence timing. *J. Dairy Sci.* 102:715–730. <https://doi.org/10.3168/jds.2018-14901>.
- Döpfer, D., A. A. H. M. ter Huurne, J. L. Cornelisse, A. J. A. M. van Assten, A. Koopmans, F. A. Meijer, Y. H. Schukken, I. Szakáll, W. Klee, and R. B. Bosma. 1997. Histological and bacteriological evaluation of digital dermatitis in cattle, with special reference to spirochaetes and *Campylobacter faecalis*. *Vet. Rec.* 140:620–623. <https://doi.org/10.1136/vr.140.24.620>.
- Egger-Danner, C., J. B. Cole, J. E. Pryce, N. Gengler, B. Heringstad, A. Bradley, and K. F. Stock. 2015. Invited review: Overview of new traits and phenotyping strategies in dairy cattle with a focus on functional traits. *Animal* 9:191–207. <https://doi.org/10.1017/S1751731114002614>.
- Egger-Danner, C., P. Nielsen, A. Fiedler, A. Müller, T. Fjeldaas, D. Döpfer, V. Daniel, C. Bergsten, G. Cramer, A. M. Christen, K. F. Stock, G. Thomas, M. Holzhauser, A. Steiner, J. Clarke, N. Capion, N. Charfeddine, J. Pryce, E. Oakes, J. Burgstaller, B. Heringstad, C. Ødegård, J. Kofler, F. Egger, and J. B. Cole. 2020. ICAR Claw Health Atlas. 2nd ed. ICAR Technical Series. ICAR (International Committee for Animal Recording). Rome, Italy.
- Engel, B., G. Bruin, G. Andre, and W. Buist. 2003. Assessment of observer performance in a subjective scoring system: Visual classification of the gait of cows. *J. Agric. Sci.* 140:317–333. <https://doi.org/10.1017/S0021859603002983>.
- Flower, F. C., and D. Weary. 2006. Effect of hoof pathologies on subjective assessments of dairy cow gait. *J. Dairy Sci.* 89:139–146. [https://doi.org/10.3168/jds.S0022-0302\(06\)72077-X](https://doi.org/10.3168/jds.S0022-0302(06)72077-X).
- Frankena, K., J. G. C. J. Somers, W. G. P. Schouten, J. V. van Stek, J. H. M. Metz, E. N. Stassen, and E. M. Graat. 2009. The effect of digital lesions and floor type on locomotion score in Dutch dairy cows. *Prev. Vet. Med.* 88:150–157. <https://doi.org/10.1016/j.prevetmed.2008.08.004>.
- Fraser, D. 2008. Understanding animal welfare. *Acta Vet. Scand.* 50(S1):S1. <https://doi.org/10.1186/1751-0147-50-S1-S1>.
- Ghotoorlar, S. M., S. Ghamsari, I. Nowrouzian, S. Ghotoorlar, and S. Ghidary. 2012. Lameness scoring system for dairy cows using force plates and artificial intelligence. *Vet. Rec.* 170:126. <https://doi.org/10.1136/vr.100429>.
- Greenough, P. R. 2007. E-Book - Bovine Laminitis and Lameness: A Hands On Approach. Elsevier Health Sciences.
- He, W., A. S. Cardoso, R. M. Hyde, M. J. Green, D. J. Scurr, R. L. Griffiths, L. V. Randall, and D.-H. Kim. 2022. Metabolic alterations in dairy cattle with lameness revealed by untargeted metabolomics of dried milk spots using direct infusion-tandem mass spectrometry and the triangulation of multiple machine learning models. *Analyst (Lond.)* 147:5537–5545. <https://doi.org/10.1039/D2AN01520J>.
- Kang, X., S. Li, Q. Li, and G. Liu. 2022. Dimension-reduced spatio-temporal network for lameness detection in dairy cows. *Comput. Electron. Agric.* 197:106922. <https://doi.org/10.1016/j.compag.2022.106922>.
- Kang, X., X. Zhang, and G. Liu. 2020. Accurate detection of lameness in dairy cattle with computer vision: A new and individualized detection strategy based on the analysis of the supporting phase. *J. Dairy Sci.* 103:10628–10638. <https://doi.org/10.3168/jds.2020-18288>.
- Klawitter, M., T. B. Braden, and K. E. Müller. 2019. Randomized clinical trial evaluating the effect of bandaging on the healing of sole ulcers in dairy cattle. *Vet. Anim. Sci.* 8:100070. <https://doi.org/10.1016/j.vas.2019.100070>.
- Knauss, M., C. L. Adams, and K. Orsel. 2022. Producer perceptions toward prevention and control of lameness in dairy cows in Alberta Canada: A thematic analysis. *Front. Vet. Sci.* 9:812710. <https://doi.org/10.3389/fvets.2022.812710>.
- Krull, A. C., J. K. Shearer, P. J. Gorden, V. L. Cooper, G. J. Phillips, and P. J. Plummer. 2014. Deep sequencing analysis reveals temporal microbiota changes associated with development of bovine digital dermatitis. *Infect. Immun.* 82:3359–3373. <https://doi.org/10.1128/IAI.02077-14>.
- Lavrova, A. I., A. Choucair, A. Palmi, K. F. Stock, M. Kammer, F. Querengässer, M. G. Doherr, K. E. Müller, and V. Belik. 2023. Leveraging accelerometer data for lameness detection in dairy cows: A longitudinal study of six farms in Germany. *Animals (Basel)* 13:3681. <https://doi.org/10.3390/ani13233681>.
- Leach, K. A., D. N. Logue, S. A. Kempson, J. E. Offer, H. E. Ternent, and J. M. Randalls. 1997. Claw lesions in dairy cattle: Development of Sole and white line haemorrhages during the first lactation. *Vet. J.* 154:215–225. [https://doi.org/10.1016/S1090-0233\(97\)80024-X](https://doi.org/10.1016/S1090-0233(97)80024-X).
- Leach, K. A., H. R. Whay, C. M. Maggs, Z. E. Barker, E. S. Paul, A. K. Bell, and D. C. J. Main. 2010. Working towards a reduction in cattle lameness: 1. Understanding barriers to lameness control on dairy farms. *Res. Vet. Sci.* 89:311–317. <https://doi.org/10.1016/j.rvsc.2010.02.014>.
- Lemmens, L., K. Schodl, B. Fuerst-Waltl, H. Schwarzenbacher, C. Egger-Danner, K. Linke, M. Suntinger, M. Phelan, M. Mayerhofer, F. Steininger, F. Papst, L. Maurer, and J. Kofler. 2023. The combined use of automated milking system and sensor data to improve detection of mild lameness in dairy cattle. *Animals (Basel)* 13:1180. <https://doi.org/10.3390/ani13071180>.
- Levac, D., H. Colquhoun, and K. K. O'Brien. 2010. Scoping studies: Advancing the methodology. *Implement. Sci.* 5:69. <https://doi.org/10.1186/1748-5908-5-69>.
- Liang, D., L. M. Arnold, C. J. Stowe, R. J. Harmon, and J. M. Bewley. 2017. Estimating US dairy clinical disease costs with a stochastic simulation model. *J. Dairy Sci.* 100:1472–1486. <https://doi.org/10.3168/jds.2016-11565>.
- Magrin, L., B. Contiero, G. Cozzi, F. Gottardo, and S. Segato. 2023. Deviation of behavioural and productive parameters in dairy cows due to a lameness event: A synthesis of reviews. *Ital. J. Anim. Sci.* 22:739–748. <https://doi.org/10.1080/1828051X.2023.2241870>.
- Manson, F. J., and J. D. Leaver. 1988. The influence of concentrate amount on locomotion and clinical lameness in dairy cattle. *Anim. Sci.* 47:185–190. <https://doi.org/10.1017/S0003356100003251>.
- Mason, W. A., E. L. Cuttance, K. R. Müller, J. N. Huxley, and R. A. Laven. 2022. Graduate Student Literature Review: A systematic review on the associations between nonsteroidal anti-inflammatory drug use at the time of diagnosis and treatment of claw horn lameness in dairy cattle and lameness scores, algometer readings, and lying times. *J. Dairy Sci.* 105:9021–9037. <https://doi.org/10.3168/jds.2022-22127>.

- McFarlane, W. J., D. L. Renaud, C. N. Reedman, T. F. Duffield, M. B. Van Camp, and C. B. Winder. 2022. A scoping review of the analytical literature concerning nonambulatory dairy cattle. *J. Dairy Sci.* 105:2544–2557. <https://doi.org/10.3168/jds.2021-21046>.
- Mellado, M., E. Saavedra, L. Gaytán, F. G. Veliz, U. Macías-Cruz, L. Avendaño-Reyes, and E. García. 2018. The effect of lameness-causing lesions on milk yield and fertility of primiparous Holstein cows in a hot environment. *Livest. Sci.* 217:8–14. <https://doi.org/10.1016/j.livsci.2018.09.008>.
- Miguel-Pacheco, G. G., H. Thomas, J. Huxley, R. Newsome, and J. Kaler. 2017. Effect of claw horn lesion type and severity at the time of treatment on outcome of lameness in dairy cows. *Vet. J.* 225:16–22. <https://doi.org/10.1016/j.tvjl.2017.04.015>.
- Mineur, A., A. Köck, C. Grelet, N. Gengler, C. Egger-Danner, and J. Sölkner. 2017. First results in the use of milk mid-infrared spectra in the detection of lameness in Austrian dairy cows. *ACS Agric. Conspec. Sci.* 82:163–166.
- Mostert, P. F., C. E. van Middelaar, I. J. M. de Boer, and E. A. M. Bokkers. 2018. The impact of foot lesions in dairy cows on greenhouse gas emissions of milk production. *Agric. Syst.* 167:206–212. <https://doi.org/10.1016/j.agry.2018.09.006>.
- Neupane, R., A. Aryal, A. Haeussermann, E. Hartung, P. Pinedo, and S. Paudyal. 2024. Evaluating machine learning algorithms to predict lameness in dairy cattle. *PLoS One* 19:e0301167. <https://doi.org/10.1371/journal.pone.0301167>.
- Noah, D. L. 2024. Basic Principles of Epidemiology - Public Health. Accessed Apr. 7, 2025. <https://www.merckvetmanual.com/public-health/principles-of-epidemiology/basic-principles-of-epidemiology>.
- O'Callaghan, K. A., P. J. Cripps, D. Y. Downham, and R. D. Murray. 2003. Subjective and objective assessment of pain and discomfort due to lameness in dairy cattle. *Anim. Welf.* 12:605–610. <https://doi.org/10.1017/S0962728600026257>.
- Ózsvári, L. 2017. Economic cost of lameness in dairy cattle herds. *J. Dairy Vet. Anim. Res.* 6. <https://doi.org/10.15406/jdvar.2017.06.00176>.
- Page, M. J., J. P. T. Higgins, G. Clayton, J. A. C. Sterne, A. Hróbjartsson, and J. Savović. 2016. Empirical evidence of study design biases in randomized trials: Systematic review of meta-epidemiological studies. *PLoS One* 11:e0159267. <https://doi.org/10.1371/journal.pone.0159267>.
- PAHO (Pan American Health Organization). 1999. Norms and standards in epidemiology: Case definitions. *Epidemiol. Bull.* 20:12–13.
- Pastell, M. E., and M. Kujala. 2007. A probabilistic neural network model for lameness detection. *J. Dairy Sci.* 90:2283–2292. <https://doi.org/10.3168/jds.2006-267>.
- Peters, M. D. J., C. M. Godfrey, H. Khalil, P. McInerney, D. Parker, and C. B. Soares. 2015. Guidance for conducting systematic scoping reviews. *Int. J. Evid. Based Healthc.* 13:141–146. <https://doi.org/10.1097/XEB.0000000000000050>.
- Post, C., C. Rietz, W. Büscher, and U. Müller. 2021. The importance of low dairy risk for the prediction of treatment events of individual dairy cows with sensor systems. *Sensors (Basel)* 21:1389. <https://doi.org/10.3390/s21041389>.
- Puerto, M. A., E. Shepley, R. I. Cue, D. Warner, J. Dubuc, and E. Vasseur. 2021. The hidden cost of disease: II. Impact of the first incidence of lameness on production and economic indicators of primiparous dairy cows. *J. Dairy Sci.* 104:7944–7955. <https://doi.org/10.3168/jds.2020-19585>.
- Dela Rue, B. T., C. R. Eastwood, J. P. Edwards, and S. Cuthbert. 2019. New Zealand dairy farmers preference investments in automation technology over decision-support technology. *Anim. Prod. Sci.* 60:133–137. <https://doi.org/10.1071/AN18566>.
- Sargeant, J. M., and A. M. O'Connor. 2020. Scoping reviews, systematic reviews, and meta-analysis: Applications in veterinary medicine. *Front. Vet. Sci.* 7:11. <https://doi.org/10.3389/fvets.2020.00011>.
- Sarker, I. H. 2021. Machine learning: algorithms, real-world applications and research directions. *SN Comput. Sci.* 2:160. <https://doi.org/10.1007/s42979-021-00592-x>.
- Schlageter-Tello, A., E. A. M. Bokkers, P. W. G. G. Koerkamp, T. Van Hertem, S. Viazzi, C. E. B. Romanini, I. Halachmi, C. Bahr, D. Berckmans, and K. Lokhorst. 2014. Manual and automatic locomotion scoring systems in dairy cows: A review. *Prev. Vet. Med.* 116:12–25. <https://doi.org/10.1016/j.prevetmed.2014.06.006>.
- Shearer, J., D. Anderson, W. Ayars, E. Belknap, S. Berry, C. Guard, K. Hoblet, E. Hovingh, G. Kirksey, A. Langill, A. Mills, D. Miskimins, J. Osterstock, R. Price, D. Prigel, A. Roussel, S. van Amstel, R. Wallace, J. Wasson, N. Cook, E. F. Garrett, D. E. Hostetler, and L. Schugel. 2004. A record keeping system for capture of lameness and foot-care information in cattle. *Bov. Pract.* 38:83–92. <https://doi.org/10.21423/bovine-vol38no1p83-92>.
- Sprecher, D. J., D. E. Hostetler, and J. B. Kaneene. 1997. A lameness scoring system that uses posture and gait to predict dairy cattle reproductive performance. *Theriogenology* 47:1179–1187. [https://doi.org/10.1016/S0093-691X\(97\)00098-8](https://doi.org/10.1016/S0093-691X(97)00098-8).
- Thomas, H. J., G. G. Miguel-Pacheco, N. J. Bollard, S. C. Archer, N. J. Bell, C. Mason, O. J. R. Maxwell, J. G. Remnant, P. Sleeman, H. R. Whay, and J. N. Huxley. 2015. Evaluation of treatments for claw horn lesions in dairy cows in a randomized controlled trial. *J. Dairy Sci.* 98:4477–4486. <https://doi.org/10.3168/jds.2014-8982>.
- Thomas, H. J., J. G. Remnant, N. J. Bollard, A. Burrows, H. R. Whay, N. J. Bell, C. Mason, and J. N. Huxley. 2016. Recovery of chronically lame dairy cows following treatment for claw horn lesions: A randomised controlled trial. *Vet. Rec.* 178:116. <https://doi.org/10.1136/vr.103394>.
- Thomsen, P. T., L. Munksgaard, and F. A. Tøgersen. 2008. Evaluation of a lameness scoring system for dairy cows. *J. Dairy Sci.* 91:119–126. <https://doi.org/10.3168/jds.2007-0496>.
- Tremblay, M., T. Bennett, and D. Döpfer. 2016. The DD Check App for prevention and control of digital dermatitis in dairy herds. *Prev. Vet. Med.* 132:1–13. <https://doi.org/10.1016/j.prevetmed.2016.07.016>.
- Tricco, A. C., E. Lillie, W. Zarin, K. K. O'Brien, H. Colquhoun, D. Levac, D. Moher, M. D. J. Peters, T. Horsley, L. Weeks, S. Hempel, E. A. Akl, C. Chang, J. McGowan, L. Stewart, L. Hartling, A. Aldcroft, M. G. Wilson, C. Garritty, S. Lewin, C. M. Godfrey, M. T. Macdonald, E. V. Langlois, K. Soares-Weiser, J. Moriarty, T. Clifford, Ö. Tunçalp, and S. E. Straus. 2018. PRISMA extension for scoping reviews (PRISMA-ScR): Checklist and Explanation. *Ann. Intern. Med.* 169:467–473. <https://doi.org/10.7326/M18-0850>.
- van der Ende, N. A., B. Roozenbeek, J. P. Broderick, P. Khatri, H. F. Lingsma, and D. W. Dippel. 2023. Blinding of outcome assessors and its association with outcome in a randomized open-label stroke trial. *Int. J. Stroke* 18:562–568. <https://doi.org/10.1177/17474930221131706>.
- van Giffen, B., D. Herhausen, and T. Fahse. 2022. Overcoming the pitfalls and perils of algorithms: A classification of machine learning biases and mitigation methods. *J. Bus. Res.* 144:93–106. <https://doi.org/10.1016/j.jbusres.2022.01.076>.
- Volkman, N., B. Kulig, S. Hoppe, J. Stracke, O. Hensel, and N. Kemper. 2021. On-farm detection of claw lesions in dairy cows based on acoustic analyses and machine learning. *J. Dairy Sci.* 104:5921–5931. <https://doi.org/10.3168/jds.2020-19206>.
- von Keyserlingk, M. A. G., A. Barrientos, K. Ito, E. Galo, and D. M. Weary. 2012. Benchmarking cow comfort on North American freestall dairies: Lameness, leg injuries, lying time, facility design, and management for high-producing Holstein dairy cows. *J. Dairy Sci.* 95:7399–7408. <https://doi.org/10.3168/jds.2012-5807>.
- Whay, H. 2002. Locomotion scoring and lameness detection in dairy cattle. *In Pract.* 24:444–449. <https://doi.org/10.1136/inpract.24.8.444>.
- Whay, H. R., D. C. J. Main, L. E. Green, and A. J. F. Webster. 2003. Assessment of the welfare of dairy cattle using animal-based measurements: Direct observations and investigation of farm records. *Vet. Rec.* 153:197–202. <https://doi.org/10.1136/vr.153.7.197>.
- Whay, H. R., A. E. Waterman, A. J. F. Webster, and J. K. O'Brien. 1998. The influence of lesion type on the duration of hyperalgesia associated with hindlimb lameness in dairy cattle. *Vet. J.* 156:23–29. [https://doi.org/10.1016/S1090-0233\(98\)80058-0](https://doi.org/10.1016/S1090-0233(98)80058-0).

- Wilson, D. J., G. Habing, C. B. Winder, and D. L. Renaud. 2023. A scoping review of neonatal calf diarrhea case definitions. *Prev. Vet. Med.* 211:105818. <https://doi.org/10.1016/j.prevetmed.2022.105818>.
- Yu, A. C., and J. Eng. 2020. One algorithm may not fit all: How selection bias affects machine learning performance. *Radiographics* 40:1932–1937. <https://doi.org/10.1148/rg.2020200040>.
- Zhao, K., M. Zhang, J. Ji, R. Zhang, and J. M. Bewley. 2023. Automatic lameness scoring of dairy cows based on the analysis of head- and back-hoof linkage features using machine learning methods. *Biosyst. Eng.* 230:424–441. <https://doi.org/10.1016/j.biosystemseng.2023.05.003>.
- Zhou, X., C. Xu, Z. Zhao, H. Wang, M. Chen, and B. Jia. 2023. Prediction of health disorders in dairy cows monitored with collar based on binary logistic analysis. *Arq. Bras. Med. Vet. Zootec.* 75:467–475. <https://doi.org/10.1590/1678-4162-12880>.

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